

HUMAN DEVELOPMENT AND COMMUNICATION

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MDC-002 INTRODUCTION TO COURSE: HUMAN DEVELOPMENT AND COMMUNICATION

Block No. 1: ‘Areas of Development Communication’ This block consists of 4 units, such as Women and Child Development, Water and Human Development, Changing Environment and Its Impact and Communication Support for Rural Development

Unit 1 Women and Child Development: This unit discusses in detail factors leading to the development of women and children are health, education, family welfare, strengthening social capital, labour, agriculture and co-operation, rural development, social security, poverty alleviation, social justice and empowerment, tribal affairs, science and technology, information and broadcasting, non-conventional energy sources, small scale and agro related industries and youth affairs and sports.

Unit 2 Water and Human Development: The present unit seeks to discuss all the significant aspects of water harvesting conservation, policy initiatives and effective ways and strategies to deal with the question of water in the 21st century in the context of development communication.

Unit 3 Changing Environment and Its Impact: Through this unit, you will learn about the changing environmental conditions and their impact. You will also know about the importance of environmental ethics and attitudes. The unit also focuses on the issues which are related to population and their consumption. In all, the unit explains the causes of changing environmental impact and in turn also categorically discusses about the impact of these changes.

Unit 4 Communication Support for Rural Development: This unit is on “Communication Support” Communication has an important role in the total development process. Communication Planning therefore, becomes an indispensable requirement and an integral part of any planning process. This unit familiarizes you with the importance of communication in rural development through an overview of communication problem in rural development, principles of communication planning, strategy, development and communication management planning.

Block No. 2: ‘Stakeholders of Development Planning’ comprises five units, Role of Public Sectors in Development, Role of Public Sectors in Development, Development Agencies, Non Government Organisation (NGO) and Panchayati Raj and Grass Root Democracy.

Unit 5 Role of Public Sector in Development: In this unit you may be able to understand the role public sector in development and the way it contributes to the idea of growth. In the first phase of independence, public sector played a crucial role in helping the core sectors to roots and insure sustained economic growth in long term.

Unit 6 Role of Private Sector in Development: like you have understood the role of public sector in the last unit the present unit will help you to understand

evolution, development and contribution of private sector in the process of national development it is very important to assess and acknowledge that in 21st century development scenario.

Unit 7 Development Agencies: This unit discusses in detail the dynamic role of various development agencies in the local, national and international context and their increasing relevance in the modern day development debate.

Unit 8 Non Government Organisation (NGO): NGO's have become household name and it is important to understand their role in the process of the development particularly in the event when government sector agencies are deciding to withdraw their control from many areas. This unit will discuss all that you need to know in this regard.

Unit 9 Panchayati Raj and Grass Root Democracy: In the discussion on development communication, it is essential to trace the growth of grassroots democracy through panchayati raj as a large section of Indian population still resides in villages. This unit will discuss the formative dimensions of the dynamics in grassroots democracy in view of its contribution to the development process.

Block No. 3: 'Media for Development' Media plays a crucial role in the development debate the present block is made of five units such as Folk and Traditional Media, Print, Radio, TV and Technological Developments in Communication. You will be able to understand the conceptual framework leading to development.

Unit 10 Folk and Traditional Media: Before the arrival of modern day media, folk and traditional media played an important role in the process of development. The present unit will take into account the role and significance of folk and traditional media so that you can understand its contribution in development communication.

Unit 11 Print: In this unit highlights the need for broad media policy framework covering the freedom of expression and the economic perspective. Then the unit explains the pre independence press, press during the freedom movement and the developments after it.

Unit 12 Radio: This unit describes the history of radio in the national as well as international context followed by sketching its political economy and growth. It also analyses the philosophy that defined radio in India.

Unit 13 TV: This unit traces the growth and history of television from both national and international perspectives, how television developed during the Gulf war and springing up twenty four hour news channels.

Unit 14 Technological Developments in Communication: This unit describes major technological revolutions that have taken place since 1430s; and links each revolution with the emergence of specific medium such as print, radio, television and digital media.

Block No. 4: 'Development Communication: Emerging Trends' This block has five units, these are Development Communication: Emerging Trends, Social & Behavioural Change communication, SBCC: Case Studies and Alternative Media.

Unit 15 Development Communication: Emgering Trends: This unit will disscuss in detail the recent and emerging trends in domain of development communication so that you can understand all the theories and perspective governing this.

Unit 16 Social & Behavioural Change communications: This unit examines the principles of SBCC and outline various strategic approaches and steps to design a SBCC strategy.

Unit 17 SBCC: Case Studies: This unit will disscuss in detail the diverse social and behavior change communication case studies with the perspective to enhance your understanding on development communication.

Unit 18 Alternative Media: This unit stresses the need for alternative media to give impetus to grassroots journalism and hightlight the issues of the marginalized sections of society.



Block**1****AREAS OF DEVELOPMENT COMMUNICATION**

UNIT 1**Women and Child Development**

UNIT 2**Water and Human Development**

UNIT 3**Changing Environment and Its Impact**

UNIT 4**Communication Support for Rural Development**

UNIT 1 WOMEN AND CHILD DEVELOPMENT

Structure

- 1.0 Introduction
- 1.1 Learning Outcomes
- 1.2 Definition: Health, Quality of Life and Anthropometry
- 1.3 Definitions: Development
- 1.4 Status of women in India
- 1.5 Status of Children in India
- 1.6 Human Development
- 1.7 Life Cycle Approach
- 1.8 Role of Media in Women and Child Development
- 1.9 Let Us Sum Up
- 1.10 Keywords
- 1.11 References and Further Readings
- 1.12 Check Your Progress: Possible Answers

1.0 INTRODUCTION

Development should be prioritised for women and children in India due to multiple reasons and factors. We need to address patriarchy and socialisation process to mainstream gender issues in the process of development. Gender relations too play a significant role in the development of male, female, children and other genders in the society. Women and men play different roles in society. Due to role differentiation, the needs of each gender differs. Society, especially Indian society, perceives girls and boys differently. The preference of boy children in the family and community is due to the same patriarchy and socialisation process. We can notice the importance of boy children through child sex ratio. The human development approach/ framework provides indicators to measure the development of children, male and female in society. By looking at the women and child development through these indicators of human development, we can understand the significance of women and children progress to make the country move forward.

As we see in this Unit, Women and children constitute more than 60 per cent of the population in India. But, in terms of all aspects of development, they are far behind male members in society. Children are also discriminated based on their sex, especially girl children discriminated, and it is difficult for them to access all facilities like male children in society in terms of health and education. We will discuss children's discrimination based on sex in the section Life Cycle approach.

Further, the public/private divide is constituted and operated in gender terms. Women expect to confine at the private sphere of life, and their existence is restricted to domestic roles predominantly. States have taken several steps after

realising the importance of women in both public and private spheres. Measures are taken from 14th Century certainly brought considerable improvements in the life of women. Socio-Political changes in the 14th Century in Europe contributed to new thinking, and it showed the path to get changes in the life of women also. Social scientists have developed a variety of theories to explain the differences persisting among human beings since the late 18th century.

At the international level, the United Nations has contributed a significant share to bringing to universal notice the status and the condition of women and incorporated women issues in all development agendas. United Nations declared 1975 as the International Women's Year. Subsequently, the United Nations organised four international conferences for women. The concept of women/child and development originated during the 1950s and the 1960s. During this period, as many as 50 countries got independence. In the year 1970, the United Nations reviewed the results of the first development decade of the 1960s. Based on the review, it facilitated to develop various approaches to women development. The study concluded that both men and women must be lifted from poverty, and both men and women must benefit from development efforts.

In the women and child development sector, governments, voluntary sectors implemented various policies and programmes which certain extent contributed to the progress of women and children, formulating laws to address their issues. The programmes include empowering strategies, employment and income generation, welfare and support services, awareness generation and gender sensitisation and other enabling measures. These programmes play the role of being both supplementary and complementary to the other general development programmes in the sectors of health, education, labour and employment and rural and urban development and so forth. Apart from programmes and policies, the government implemented various support structures through voluntary agencies to assist in the development of the women and children. The main areas that focused upon in leading to the development of women and children are health, education, family welfare, strengthening social capital, labour, agriculture and co-operation, rural development, social security, poverty alleviation, social justice and empowerment, tribal affairs, science and technology, information and broadcasting, non-conventional energy sources, small scale and agro related industries and youth affairs and sports (Women and Children, 2007).

1.1 LEARNING OUTCOMES

After studying the Unit, you would be able to:

- Define health, Quality of life and anthropometry;
- Discuss the status of women and children;
- Explain the status of women and children through the human development approach and Life cycle approach; and
- Examine the role of media for the development of women and children.

1.2 DEFINITION: HEALTH, QUALITY OF LIFE AND ANTHROPOMETRY

According to World Health Organization (WHO), 'the enjoyment of the higher attainable standard of health is one of the fundamental rights of every human

being”. According to Amartya Sen, ‘health is among the most important conditions of human life and a critically significant constituent of human capabilities which we have reason to value’ (2002). In addition to the intrinsic value for Health, it is also significant aspects to achieve economic growth, educational attainment, succeeding in employment, earning income, travel, maintaining a social life, safety, security and empowerment all especially women and children. According to Constitutional Preamble of WHO, ‘Health is a state of complete physical, mental and social well-being. It is not merely the absence of disease or infirmity. This definition was adopted at the International Health Conference, New York, 19-22 June, 1946; signed on 22 July 1946 by the representatives of 61 States (Official Records of the WHO, no.2, p. 100) and entered into force on 7 April 1948.

Health has various components- physical, mental, emotional, social and sexual with the following characteristics:

- Each component affects the distinct aspect of Health.
- All components are interrelated.
- All components together determine the Health of an individual.

We can conclude that Health is a complete well being of individuals and children by looking at the above definition. In the following sub-section, let us try to understand another significant concept related to Health, that is quality of life.

Quality of Life (QoL)

The concept has a different meaning in philosophical, political and Health related contexts. Health-related QoL (HRQoL) includes physical, functional, social and emotional well-being of an individual. Miles defined Quality of Life (QoL) as an aggregate of several social and economic outputs received by groups or individuals or as subjective evaluation of satisfaction. It is also called as ‘well-being’. The following small study conducted by Parthasarathy among Irula tribes in Niligiri hills gives us clarity on the definition ‘Quality of Life’. In his research, he described the life of the Irula tribes. They lead a simple life by making themselves satisfied with practical needs.

They use locally available goods and services such as food, shelter and drinking water. However, outsiders have different perceptions by looking at the life of Irula tribes. Members of the tribes are not comfortable with outsiders’ interference and criticism about their way of life and their socio-cultural needs. They request outsiders not to observe their way of living. They ask others to stay away from them. This study shows that Irula tribes perceive the quality of life based on the socio-economic outputs of their community. They content with their life. But, outsiders view them subjectively and consider their life as not a ‘quality life’ according to their perception. According to UNESCO (1977) QoL is as an inclusive concept which includes all aspects of life, both the material needs and intangible factors like personality development, self-realization and healthy ecosystem. QoL is operating at both the micro and macro level. Quality of life at the micro-level includes food, shelter, clothing, drinking water and disposable resources. Quality of life at macro level includes infrastructure facilities, transportation, the safety of individuals and responsive governance. We can measure QoL through indicators.

Anthropometry

Apart from the definition as mentioned above, other significant indicators to measure child health is the measures of anthropometry that is the measure of height for the age, weight for height (Body Mass Index). These aspects will explicitly talk about nutritional aspects among children. Poverty will undoubtedly help us to know both children and adults, especially women's quality of life and the measures to be taken through appropriate policies for the improvement of life of women and children. Stunted growth among children will affect cognitive development and educational attainment. Other significant indicators which show the Health of the children and women are Infant Mortality Rate (IMR) Child Mortality Rate (CMR), Maternal Mortality Rate (MMR), Iron and Folic acid deficiency, access to health care facilities, Vaccinations, trained attendants.

1.3 DEFINITIONS: DEVELOPMENT

The growth and Development of women and children are significant for the holistic growth of the country. Efforts have been taken at the International and national levels to mainstream Development of women and children through the formulation of appropriate policies and programmes. India consists of 121.06 crore population as per 2011 census. The female population is 48.5% out of the total population. Out of the total population, the percentage of children's population (0-6years) is 13.12 per cent. Thus, women and children consist of 62.62 per cent as per 2011 census. The Development of a large section of the population in India is imperative to make the country developed. The term 'development' perceived and defined differently by academic and policymakers. There are two approaches used in this Unit to learn about women and child development. One is a human development approach which provided various indicators to measure the Development of groups. The second one is Life cycle approach. Before we proceed further to see the Development of women and children through HDI (Human Development Index) and GDI (Gender-Related Development Index) and GEM (Gender Empowerment Measures) indicators, let us learn the term 'Development.'

Development has many meanings. Regardless of its definition and the framework, Development as a multidimensional and multi-sectoral process, involving social, economic and political change aimed at improving people's lives' (refer). The process of Development manages and utilizes natural resources to meet the needs of human beings. Many accept this view irrespective of their ideological positions. Generally, some regions considered developed due to rapid economic activities, possessing high per capita income by the inhabitants, lots of employment activities and consumption of goods and services. Here economic growth and productive investment are the primary concerns. Another development perspective provides importance about people's health and education, which are significant to produce a more dynamic economy, a healthy and prosperous society. Development should also enable people to live the life they perceive and value. In this perspective, investment, economic activities, infrastructure development and employment opportunities are mean to achieve the Development in the areas health, education, child development, women empowerment. The human development approach addresses all these aspects.

1.3 STATUS OF WOMEN IN INDIA

Status is a state, condition, or situation. Sociologically speaking, Social status is the honour or prestige attached to person's position in society (one's social position). It may also refer to a rank or position that one holds in a group. Social status, the position or rank of a person or group within the society, can be determined in two ways. One can earn their social status by their achievements, which is known as achieved status. Alternatively, one can be placed in the stratification system by their inherited position, which is called ascribed status. Ascribed statuses can also be defined as those that are fixed for an individual at his/her birth.

Ascribed statuses that exist in all societies include those based upon sex, age, race, ethnic group and family background. For example, a person born into a wealthy family characterized by traits such as popularity, talents and high values will have many expectations while growing up. Therefore, they are given and taught many social roles as they are socially positioned into a family. Society expects them to become equipped with all traits and characteristics by acquiring through training and education. Achieved statuses meaning also what the individual acquires during his or her lifetime as a result of the exercise of knowledge, ability, skill and/or perseverance. Occupation provides an example of status that may be either ascribed or achieved, it can achieve by one gaining the right knowledge and skill to become socially positioned into a higher position of that job; building a person's social identity within the occupation. We will look at the status and position of women in our society in various fields based on the above description.

In India, Sex ratio is defined as the number of females per 1000 males in the population. In contrast, in almost all the UN publications/International publications, it is expressed as males to per 100 females. During the post-Independence period from 1951 to 2011, sex ratio in rural India has decreased from 965 to 946 and increased from 860 to 929 in urban India. At all India level, the sex ratio had decreased from 946 in 1951 to 943 in 2011. During this period 19 States/UTs have recorded a significant increase in sex ratio. The notable increase has been recorded in Andaman & Nicobar Islands, (40.16%), Delhi (13.02%), Assam (10.34%) and West Bengal (9.82%). Contrary to this, 16 States/UTs have recorded a significant decrease in Sex ratio. Notable reduction in sex ratio has been recorded in Daman & Diu (45.03%), Dadra & Nagar Haveli (18.19%), Goa (13.71%), Lakshadweep (9.25%) and Bihar (8.21%).

Table 1.1: Gender inequalities- Some facts in India

Demographic Profile				
Population	Unit	1991	2001	2011
Total	Crore	84.6	102.9	121.1
Female	Crore	40.7	49.6	58.7
Male	Crore	43.9	53.2	62.3
Sex Ratio				
All India		926	933	943
Rural		938	946	949

Urban		893	900	929
Life expectancy	Years	2001-05	2006-10	2011-15
Male		63.1	64.6	67.3
Female		65.6	67.7	69.6
Literacy rate (7+ years)	%	1991	2001	2011
All		52.21	64.84	72.99
Male		64.13	75.26	82.14
Female		39.29	53.67	65.46
Higher Education	%	1950-51	2005-06	2013-14
Total		N.A	11.16	21.1*
Male		N.A	13.5	22.3
Female		N.A	9.4	19.3

Reference: India Figures 2015, Ministry of Statistics and Programme Implementation

***provisional**

The evidence is shown in the above table that gender differences and inequalities persist at all levels and areas. According to Oxfam study, Gender inequalities is one of the world's oldest and pervasive disparities, and it is continuing till now. As of 2013, an estimated 767 million people lived below the international poverty line of \$1.90 a day (Progress towards the Sustainable Development Goals (E/2017/66. UN Women). UN Women and World Bank analysed the existing poverty data to submit in the special event of the Economic and Financial Committee of the UN General Assembly on 18th October, 2017 and found the gender differences in Poverty.

According to Global Micro Database (GMD) presented by the Kinnin Scott of the senior economist in the world Bank, women are more likely to be poor than men between the age of 20 to 34 years. Women are likely to be poor in their productive age than men. Poverty, combined with low education makes women to join in the unorganised sector. Women may accept to marry early in their period and produce children due to poor education, lack of access to resources and lack of access to health facilities at an early age. Children born for young, poor women may possess poor health like low weight, stunted growth and nutrition deficiency. Divorce, separation and widowhood also affect women more negatively than men. Divorced women in the 18-49 age group are more than twice as likely to be poor than divorced men in that same age group.

The research also showed that the poorest, single parents with children, and mothers with children might face a far higher risk of Poverty. Governments must consider all these factors before formulating policies to address poverty and inequalities.

“The greater likelihood of women not having an income of their own and the inequality in the division of unpaid care work, put women at a severe economic disadvantage compared to men, and at a higher risk of poverty,” says Shahra Razavi, Chief of Research and Data Section at UN Women. “

The evidence from Esther Boserup work on African Agriculture pattern in her book “Women’s Role in Economic Development” clearly brought out the women’s involvement in food production. But ownership of land is with men in most of the societies according to the Food and Agriculture Organization (FAO).

The World Economic Forum’s Global Gender Gap Index measures the extent of gender-based gaps in economic participation and opportunity, educational attainment, health and survival, and political empowerment. India ranked 112th position in 2019-2020, four places below from 2018 rank. India was 98th position in Global Gender Gap index 2006, and it has slipped in 112 positions in 2019-2020. India was 103 position with the score 0.962 2006 in health and survival. India has again moved backwards, and it has become the 150th position with the score 0.944. The Gender Gap Index assesses countries on how well they are dividing their resources and opportunities among their male and female populations, regardless of the overall levels of these resources and opportunities.

By providing a coherent framework for assessing and comparing global gender gaps and by revealing those countries that are role models in dividing these resources equitably between women and men, the Index catalyzes greater awareness as well as the greater exchange between policymakers. India is arguably amongst the most complex democracies in the world, with a long history of commitment to women’s welfare and empowerment through Constitutional provisions, legislation, policies and plans even while deep levels of structural gender bias persist within families, societies and economies. There are challenging opportunities and challenges, including continued high rates of economic growth in the face of a declining sex ratio, decreasing energy and natural resources, increasing income inequality and an urban-rural divide.

An informed and purposive approach to addressing the inequities of women’s social, political and economic exclusion, particularly amongst excluded groups, can no longer be viewed in terms of women’s issues. Instead, gender equality is a core prerequisite for sustainable and equitable economic growth. No discussion on women, gender, planning and policy can happen without addressing the context in which Indian women live and die. The links between women’s economic empowerment and the declining sex ratio are more important than at first glance. The constant undervaluing of women’s worth in economic and social terms and statistical ability to measure the differences has brought the issue to ahead.

1.5 STATUS OF CHILDREN IN INDIA

The life of the child starts in the mother’s womb. Once a woman conceives a baby, society expects a male child. If a woman delivers a girl baby, social institutions like family start discriminating women. Family start neglects the health of the mother and child. If a woman has a male baby, the family provides nutritious and sufficient food to a male child. Among Children, girls may face discrimination in the early stage. Family tends to discriminate adult family members too. Male members in the family get preference in getting nutritious and sufficient food and female members in the family likely to consume left over meals. According to a report by the United Nations Children’s Emergency Fund (UNICEF) in the year 2011 on **Status of Children in India A profile**, prevalence of anaemia among currently married women and girls in the age group of 15-19 years is very high. About 56 per cent of recently married women and girls in the

age-group 15-19 years have a lower haemoglobin level than 12 g/dl. Once a child grows up, the family provides education based on child sex. The concept of breadwinner plays a significant role in the education of the child. The male child expects to study well, and he is supposed to take care of the entire family economically. Thus, the family tends to provide quality education to boys, and girls can have an education if it is accessible and affordable. Due to discrimination in the early education, Girls grow up with less skilled and ended up joining in semi-skilled or low professional like the informal economy. They tend to earn less than their counterparts. It further reflects in every aspect. Low level of education affects their freedom to decide occupation, marriage, and it restricts women's mobility. It also impacts their sexuality and reproductive capacity.

The patriarchy and socialisation process decide the growth of the children. Society already perceives about children based on sex. Social institutions try to make everyone believe that a girl child should be brought in a particular manner, and boys should be brought in a certain way. This natural and individual cannot bring changes in the existing gender roles. Restricting the mobility of women and girl children particular way affect the growth and development of the girls. Low education affects women's capacity to decide on their own. It may reflect every aspect of their life. Low educated, unskilled women may accept to get married early in their life. Early marriage, lack of proper nutrition, conceiving a child at an early age may produce unnourished /malnourished child. This becomes a vicious cycle, and it may continue till women get proper education and get the capacity to decide their life on their own.

The life cycle of women and the problem faced by women in each life cycle period has been discussed in the following section. In India, when we look at the health indicators like Maternal Mortality Rate (MMR), Infant Mortality Rate (IMR), sex ratio, life expectancy at birth of women, the numbers are high compared to other developing countries. Apart from personal changes, it devalues their significance in the family and society.

In the following section, you will study the **life cycle approach** and how discrimination happens at each stage of women's life. We will discuss the life cycle approach in detail later in this Unit with more information.

1.6 HUMAN DEVELOPMENT

In this section, we will discuss women and child development from the human development perspective. Human Development approach provides people as a priority for development. The economy also important, but it enables people to enjoy full freedom, and it gives a choice to the people to pursue what they consider as development. According to this approach, a healthy economy must be created with necessary opportunities for the people to enjoy a healthy and long life with educational opportunities, a meaningful job, physical safety and democratic debate. Unlike other development approaches, Human development approach gives priorities to both male and female and children. Many focused on people's lives, still, they consider economic growth, macroeconomic stability, income and poverty reduction and improving people's lives are their priority areas—others who concentrate on sustained growth concern themselves with healthy, educated society with peace and stability. The second approach is based on human development approach. It also referred as human development and capability approach. It is focused on all peoples in the society and their freedoms. The notion of people's freedom can be found in the idea of Ubuntu in Southern Africa,

with liberation theology in Latin America and beyond with participatory development, community mobilisation, rights-based approaches, sustainable livelihoods and many other ethical approaches to development. It is applicable every country irrespective of whether they are developed or developing. Many perceived, theorised and developed indicators to measure human development in the society.

Nobel Laureate Amartya Sen's theory of capability approach provided the philosophical basis of human development. We need to note the contribution of Mehboob Ul Haq and Martha Nussbaum. The development of women and children are measured through various indicators like Sex ratio, Infant Mortality Rate (IMR), Maternal Mortality rate (MMR) so on, so forth as per human development approach.

Check Your Progress: 1

Note: 1) Use the space below for your answers.

2) Compare your answers with those given at the end of this Unit.

1) What is Human Development Approach?

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.....

1.7 LIFE CYCLE APPROACH

This section is adapted and modified from the following website. http://www.businessballs.com/erik_erikson_psychosocial_theory.htm#erikson's_basic_virtues

Empowering women as a process, and it requires a life-cycle method. Therefore, every phase of their life counts as primacy in the planning process. Depending upon the progressive requirements at every stage, the female population has been characterized into five distinct sub-groups as the population anticipated for 2001 (Women and Children, 2007).

Human beings are born as infants and slowly pass through different stages and become adults. These stages constitute the life cycle. A life cycle is defined as the developmental stages that occur during an organism's lifetime. A life cycle ends when an organism dies.

For example, let us see the life cycle of plants. The life cycle of plants starts as seeds, and it sprouts as seedlings. Seedlings grow into offshoots and then become mature plants. Mature plants produce seeds. From these seeds life restarts. A life cycle may comprise more than 3 primary stages depending on the species. The exact names of each stage can also vary slightly depending on the species. Erik Erikson, a famous psychologist first published his eight-stage theory of human development in 1950 in his book "Childhood and Society". The chapter featuring the model is titled 'The Eight stages of Man'. Erikson developed his theory mainly from his extensive practical field research.

Erikson's eight-stage theory is relevant even in the present context. He explains how personality and behaviour develop in people. Erikson's theory is useful for teaching, parenting, self-awareness, managing and coaching, dealing with conflict, and general understanding self and others.

The table below shows Erickson's eight stages of 'psychosocial crisis stages', age at each stage and life stage.

Table 1.2: Erikson's eight psychosocial stages

Psycho- social Crisis Stage	Life Stage	age range, (other descriptions)
1. Trust v Mistrust	Infancy	0-1½ yrs baby (birth to walking)
2. Autonomy v Shame and Doubt	Early Childhood	1-3 yrs (toddler, toilet training)
3. Initiative v Guilt	Play Age	3-6 yrs (pre-school, nursery)
4. Industry v Inferiority	School Age	5-12 yrs (early school)
5. Identity v Role Confusion	Adolescence	9-18 yrs (puberty, teens*)
6. Intimacy v Isolation	Young Adult	18-40(courting,early parenthood)
7. Generativity v Stagnation	Adulthood	30-65 (middle age, parenting)
8. Integrity v Despair	Mature Age	50+ (old age, grandparents)

Ref:http://www.businessballs.com/erik_erikson_psychosocial_theory.htm#erikson's_basic_virtues

Erickson believes that the psychosocial crisis mentioned in each stage affects human beings' development and personality. He said both positive and negative psychosocial situation in each stage. He derived the term 'psychosocial' from two words namely psychological (or the root, 'psycho' relating to the mind, brain, personality, etc.) and social (external relationships and environment). Erickson believes that his psychosocial principle is genetically inevitable in shaping human development occurring in all people. Erickson believes in nurturing of human personality and behaviour.

The term psychosocial is important in Erickson's theory. As we have seen in the table, there are three columns. Erickson mentioned two words in column 1, namely 'psychosocial crisis'. He said the word Trust Vs Mistrust during the infancy period. Successfully passing through each crisis in a particular stage is significant. The balance between trust versus mistrust helps human beings to pass through stage healthily. So, Erickson has given two words in each stage. One is positive, and the other is negative. These words are symbolic. Human beings may pass through both negative and positive experiences in each life stage. This does not mean that if someone goes through a high percentage of negativity in one stage, she/he cannot come out. One can quickly come out from negativity if we consciously pass through it. After looking at Erickson's stages of human development, let us see how women face a crisis in each stage.

Infancy - (0-1 yrs 6 months): Once the family and society come to know about the birth of the girl child, they commit female infanticide, and such cases are reported in many parts of the country. Even if the girl child survives, they may get discriminated viz., without giving nutritious food and asking mother not to feed girl child properly. Before birth, sex-selective abortion is prevalent. Due to sex-selective abortion, the sex ratio is low in India. The following table gives you the sex ratio of select few states in India.

Table 1.3: Sex ratio in a few States of India

Sl.No.	States	Sex Ratio	Child Sex Ratio
1.	Kerala	1084(highest)	964
2.	Puducherry	1037	967
3.	Tamil Nadu	996	943
4.	Andhra Pradesh	993	939
5.	Chattishgarh	991	969 (highest)
6.	Haryana	879	834
7.	Andaman and Nicobar Islands	876	968
8.	Delhi	868	871(lowest)
9.	Chandigarh	818	880
10.	Dadra Nagar Haveli	774(lowest)	926

Computed: Census of India, 2011

Early Childhood (toddler, and toilet training)- 1-3 yrs: Like infant stages, girls are discriminated based on food; clothing and the socialisation process in this AgeAge too.

Play Age (pre-school, nursery) -3-6 yrs: Family may not prefer to send their girl child to play school. They may get discriminated for food, clothing and medical care.

School Age (early school) 5 -12 yrs: Girls are socialised to take care of the younger siblings at this stage. If there is more work at home, girls are asked to discontinue their studies. In India, children (girls and boys) complete primary schooling at the AgeAge of 10. In many parts of the country, the upper primary, high and higher secondary school may not be available in a closer vicinity. In that case, the family refused to send a girl child for schooling to faraway school. Lack of education can have a severe impact on the decision making of the girl, including on the health front. Girl children also face a lack of medical aid. They may also face physical violence.

Adolescence (puberty, teens ages) -9-18 yrs: Concerning a girl child, this is the Age for attaining puberty. They need psychosocial counselling and medical attention to understand the puberty. Family and society do not address their need. Schools, communities and governments at the local level must help the girls to understand physical and physiological changes in their body.

Young Adult (courting, early parenthood) 18-40: As an adult and adolescent women, they may face the problem of early marriage, withdrawal of support to continue education, early and repeated pregnancy, sexual violence and domestic violence. Maternal health remains a grave concern at this stage. The common cause for maternal deaths in India is haemorrhage, abortion, infection, obstructed labour, eclampsia (blood pressure during pregnancy), sepsis and anaemia.

Adulthood (30-65): Domestic and sexual violence, stress, triple burden are some of the issues women face during this period.

Mature age (old age, grandparents) 50 +: Like the previous stage, violence is the major problem women face in this stage too. The family may abandon woman again and may force to leave home.

The following two case studies will help you to appreciate government initiatives of two State governments in India to address the health concerns of all, mostly women and children. The Case study Tamil Nadu shows the overall improvement in public health. In this process, women and children are benefitted, and essential health indicators like Maternal Mortality Rate (MMR), Infant Mortality Rate (IMR) and Child Mortality Rate (CMR) have been reduced.

The following case may provide better understanding about child development in her/his early life. The case study is developed and published by the UNICEF. We have adopted the case study to make our learners to understand the gravity of the situation at the grassroots from their website. The link is given below.

<https://www.unicef.org/india/stories/memories-stillborn-child>

08 October 2020

"I don't blame anyone. The nurses in the hospital said I should have come earlier. But what has happened has happened," says Golapi (33), reminiscing her grief of giving birth to a stillborn baby boy through caesarean section. She does not want to relive those days of trauma and pain that followed when her third child was born stillbirth. Golapi never got a chance to go to school. Her life in the remote Jambahali village of Nunpur Panchayat in Kalahandi district, Odisha, rural India, is a mix of household work, occasional farming in the fields and taking care of her husband Bashistha Rana and two children – a 15-year-old daughter and six-year-old son.

Golapi had been diligently following the instructions of the village health worker and had been regularly going for her antenatal check-up visits. In early June 2020, the auxiliary nurse midwife (ANM) had identified hypertension during her pregnancy as a "high risk" condition and had referred her to the nearest community health centre, 30 kilometres away in Tusra, for further check-up. Her family, however, took her to the district headquarter hospital (DHH) in Bolangir. She had given birth to her first child at the DHH and had planned the delivery of her third child too there. She was given medicines for high blood pressure and an ultrasound revealed that the baby was fine and weighed around 3.2 kilograms. After a four-day stay at the hospital, Golapi returned home.

However, a week later, her placenta gave way and she could feel that the baby was not moving. The decision to go to the hospital, followed by locating and hiring a vehicle took more than two hours. The couple set out to the community health centre in Tusra, which took more than an hour to reach. The doctors there referred her to DHH Bolangir, where she delivered by a caesarean section and gave birth to a stillborn baby. The doctors explained to her that she "should have come earlier."

But how early could Golapi have come? The nearest community health centre is 30 kilometres away in Tusra and the closest large, well-equipped hospital – the District Headquarter Hospital – is 70 kilometres away in Bolangir from Jambahali. The road conditions are low, and it takes more than three hours to reach Bolangir.

High blood pressure leading to separation of the placenta in the womb was the most likely cause of stillbirth and could have been prevented if Golapi was able to reach the hospital on time. It could have helped manage her condition of pre-eclampsia leading to separation of the placenta in the womb.

Socially, the reasons for stillbirth can be traced back to her pregnancy as an adolescent. Early pregnancies usually leave life-long impacts.

India committed to bringing down stillbirth rate.

“India has made considerable progress in reducing stillbirths in the past few years. It is heartening to see the high-level of political commitment and investment towards ending preventable stillbirth deaths in the country. Every stillborn baby is a tragedy. A vast majority of stillbirths can be prevented through a strong primary health care system. It is our joint responsibility to ensure that every woman is aware of, and adopts healthy behaviours and practices during pregnancy. It is critical that women receive timely access to skilled and quality care before and during childbirth,” says Dr Yasmin Ali Haque, UNICEF India Representative.

Pre-existing health conditions, adolescent pregnancies, infections during pregnancy, high rates of anaemia among pregnant mothers, conditions like pre-eclampsia, inadequate access to quality services and poverty are some known factors that contribute to the high rate of stillbirths. A majority of the cases occur during delivery or during the last weeks of pregnancy. India has committed to reduce further the number of stillbirths, as part of its agenda to improve maternal and newborn health.

Arti Senapati, an auxiliary nurse midwife (ANM), who has helped countless women in rural Odisha to have safe pregnancies over the last 32 years, says, “I know of women who get to eat only rice with a small portion of onions, chillies or a few pieces of vegetables. Then there are those, who continue to do hard labour even during the advanced stage of pregnancy. Early marriages and frequent pregnancies also increase the risk of stillbirths.”

Over the three decades of her career, Arti has seen progress in the number of stillbirths coming down. “There have been several initiatives by the government and more women today are aware that they need to go to the local health worker during their pregnancy, take Iron Folic Acid supplements and take the mandatory immunization shots. I also see women wanting to report to the hospital before their due date,” she adds.

“Birth spacing to prevent frequent pregnancies also needs a bigger push. Women in tribal communities also need more support as they report several health issues during pregnancy,” Arti says.

Golapi, since, has undergone tubectomy. “I already have two children and do not want any more,” she says. While she may have reconciled with the difficult experience and does “not want to relive the trauma,” there are many women who can’t forego the memories of their pregnancy and the trauma of the loss of their child.

As an ANM and a mother herself, Arti understands the deep pain women undergo post stillbirth. She stresses the need to be by the mother’s side and provide her

and the family emotional support. "I remember a young tribal woman who lost her first child and was traumatized by the grief. I stayed in regular touch with her and counselled her. Later she gave birth to a healthy child," she says.

The above case study clearly taught us the significance of marrying at right age, proper nutrition, accessing health facilities, accessible and affordable health services, nutritious meal for pregnant women and infant to make our society healthy and developed.

1.8 ROLE OF MEDIA IN WOMEN AND CHILD DEVELOPMENT

Mass Media play a significant role in bringing awareness about development issues. In the era of neo-liberal economy and getting accessed smart phones is affordable. Thus, mass media, including content streamed through smartphones through phone apps, can be used in both positive and negative manner. It is significant to use mass media positively to make society a better place for all. Mass media may negatively portray women due to patriarchy and socialisation process. It is significant to understand the existing gender socialisation to depict women and children in a dignified manner to bring development among them on par with male members in society.

The status of women and girls within the society should be acknowledged. Secondly, National Crime Record Bureau (NCRB) publishes data on crimes against women in various sections regularly. Not only publishing the data by the media, but it should also further analyse the data, and periodically publish to bring more extensive awareness among masses on gender-based violence (GBV) in the society. Media must stop publishing stories, which explicitly or implicitly demeaning, degrading and negative conventional stereotypical images of women. Private players control a significant percentage of mass media. It must make sure that women can also access equally like male members at all levels in the media industry. Media houses can design and develop gender-based code of conducts, professional guidelines to ensure the entire space is gender-sensitive.

It must provide compulsory gender sensitisation training to all to remove gender stereotypes, and it must encourage to publish development news and views to bring development for all. As we have seen in the human development approach, development of women measured by accounting disparities between male and female in the select indicators. United Nations publish World Human development report every year and we can see the country's position and progress through these reports. The media need to highlight the position of the country widely to make all stakeholders sensitize on the significance of developing all along with economic development.

Secondly, we have seen the life cycle approach in this Unit. Children especially girl child and women, suffers every stage of their life. It starts with the birth of a girl child. One can understand the discrimination of infant based on sex, violence against women and girls through the data in this Unit. Government publishes gender-disaggregated data at the state level and district level regularly. It is the responsibility of the media to highlight the women and children's status and position through these indicators. It is also essential to analyse the data and highlight the causes and consequences along with best practices.

Violence against women is pressing issues. All forms of violence against women, physical and psychological, whether at the domestic or communal levels, including those arising from practices, rituals or accepted beliefs should be published regularly with gender-sensitive manner to disregard these incidences. Rights of the Girl Child - All forms of discrimination against the girl child and violation of her rights should be eliminated by undertaking strong procedures, both preventive and punitive within and outside the family. These would communicate precisely to the society by the mass media.

1.9 LET US SUM UP

We have seen the development of women and child from the perspective of human development and life cycle approach. It is essential to bring development among women and child on a par with their counterparts, male members. We understood women and child development through development indicators. We have concentrated the development by looking at the health data. If we further expand our analyses, they still far behind male members in society. Thus, creating opportunities and expanding choices for women and children is vital to make them developed.

1.10 KEYWORDS

Patriarchy: This term “patriarchy” derives from nineteenth century anthropology and literally means ‘the rule of the father’ (from latin, pater is father and arche is rule).

Gender Socialisation: It is the process of learning societal expectations, attitudes and behaviours associated with boys and girls. Once a child born, they society expects social institutions like family to bring a child in a way society expects.

Maternal Mortality Rate: Maternal Mortality Ratio is number of women who die from pregnancy related causes while pregnant or within 42 days of pregnancy termination per 100000 live births.

Infant Mortality Rate: Number of infant deaths in a single year out of every 1,000 live births that year

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1.12 CHECK YOUR PROGRESS: POSSIBLE ANSWERS

Check Your Progress: 1

1. It provides people for developments please refer 1.6 for elaborate answer.

UNIT 2 WATER FOR HUMAN DEVELOPMENT

Structure

- 2.0 Introduction
- 2.1 Learning Outcomes
- 2.2 Water and Development: An Overview
- 2.3 Water as a Sustainable Development Goal
- 2.4 Water: Issues and Challenges
 - 2.4.1 Water Budgeting
 - 2.4.2 Water Usages and Demand
 - 2.4.3 Significance of Rainwater harvesting and Conservation
 - 2.4.4 Water Quality Deterioration
- 2.5 Major Water Government by GOI
- 2.6 Water and Agriculture
 - 2.6.1 Irrigation Method
 - 2.6.2 Advance Irrigation Methods for Water Conservation
- 2.7 Water for People's Well Being
- 2.8 Water for Economic Development
- 2.9 Let Us Sum Up
- 2.10 Key Words
- 2.11 References and Further Readings
- 2.12 Check Your Progress: Possible Answers

2.0 INTRODUCTION

Water is an important natural resource required for survival of human kind and has a significant impact on human development. The role of water in human development cannot be underestimated as it required for each and every aspect of people's lives and livelihoods. The ever-increasing population growth and urbanization creates pressure on water resources resulting in threat to climate change affecting the complex relationship between water and human beings. Water and human development are closely linked. According to United Nations (UN) General Assembly, safe and clean drinking water has been a Human Right since 2010. Water is unquestionably having social significance as it plays an important role in social dynamics, economics and overall development.

Water assumes a central role in the policy debates and it is increasingly argued that the growth and development of a society and nation is determined by the level of awareness and sensitization about water. The present unit discuss about the significant aspects of water harvesting conservation, policy initiatives and effective ways and initiatives to deal with the question of water and sustainable development in the 21st century in the context of development communication.

2.1 LEARNING OUTCOMES

After studying this unit, you should be able to:

- understand the role and significance of water in the development debate;

- explain water usage and demand;
- acquaint yourself with water in respect to rainwater harvesting; and
- describe the SDG of water and impact on global policy perspective in the context of development communication.

2.2 WATER AND DEVELOPMENT: AN OVERVIEW

Without water, the living earth cannot be; if so for anyone, without rain order cannot be.

(Thirukkural, 2.10)

Water is a critical input as a natural resource for every form of life in the Universe. Water plays a vital role in each and every aspect of human and nation development such as socio-economic development as well as maintenance of healthy ecosystems. Water is the principal natural resource and covers almost two third earth's surface. Only 3% of water is freshwater, of which just one third is accessible for human use that may be for domestic, industry or agriculture.

The importance of water cannot be ignored with the evaluation of this planet and it is a fact that almost all the civilizations move around water i.e. appeared on the banks of the big rivers. The importance of water is mentioned in all religious literature. Water is one of the five *panchmahabhutas* or great elements of life even in the ancient Indian tradition. Early Indian literature belonging to different religious literature including Hinduism, Buddhism, Jainism, and other traditions had highlighted the importance of water and its conservation. In Ayurvedic Literature water is defined as *jiva* (life) and has countless treatments on water. However, this tonic of life is becoming increasingly scarce due to different reasons.

The rapid increase in population, growing urbanization, industrialization, change in lifestyle, water misuse, and globalization have led to severe crisis of usable water in the world, particularly in developing countries like India. The annual per capita water availability has shown an unswerving declining trend over the years and it was 5200 m³ in 1951, 1545 m³ in 2011, may further reduce to 1486 m³ in 2021 and it is estimated to decline further upto 1140 m³ in the year 2050. There is a substantial gap between the demand-supply since the requirement for water in various sectors is continuously increasing and in turn the resources are rapidly depleting. Further, demand for water from various sectors *viz.* irrigation, drinking water, industry, energy and others are expected to rise from 710 billion cubic metre (BCM; Billion (109) Cubic Meters = 103 MCM=1Km³) in the year 2010 to 843 BCM in the year 2025 and further to 1180 BCM in the year 2050. According to a 2018 NITI Aayog Report, almost two lakh people die every year due to inadequate access to safe, good quality potable water and presently about 60 crore citizens of the country face high to extreme water stress conditions.

There is an urgent need for adopting water harvesting and conservation measures like rainwater harvesting, preventing pollution, management of surface and groundwater resources etc. It is envisaged that active participation of local people plays a pivotal role in management of this important natural resource. In India, annual per capita water availability is less than the threshold value of 1700 m³ and being declared as a water scarce region of the world as shown in Fig. 2.1. When annual per capita renewable fresh water availability is less than 1700 m³, periodic or regular water stress is experienced. The projected annual per capita

water availability in 2025 and 2050 is likely to be 1401 and 1140 m³, respectively which is very close to the level of 1000 m³. At this level, water scarcity may begin to affect economic development and human health.

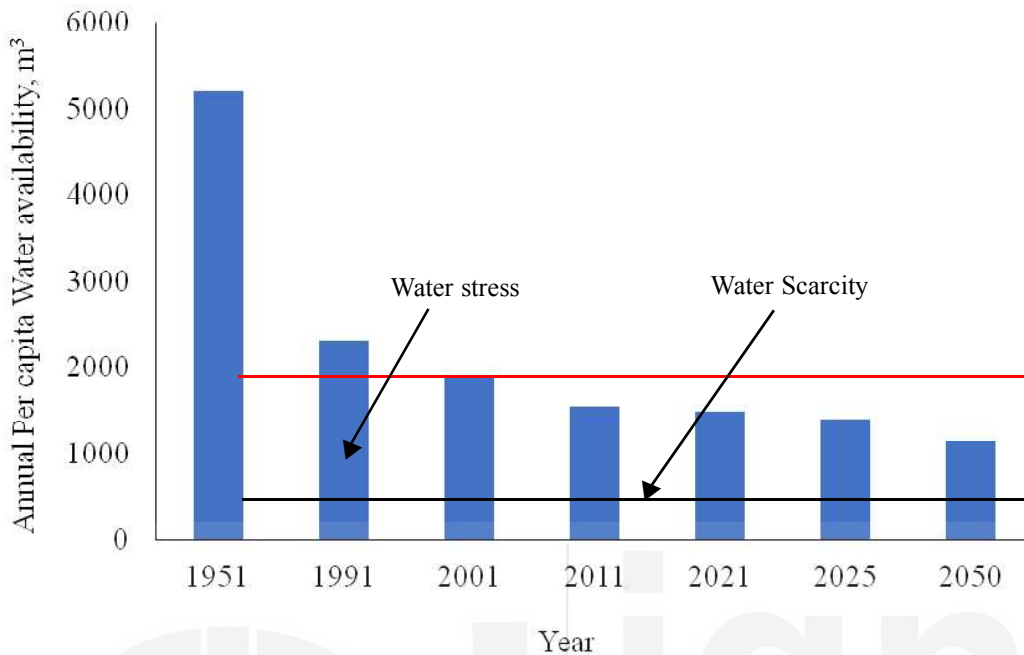


Fig. 2.1: Present and projected per capita annual water availability in India

Water is a critical input for survival of life on this universe as an input for agriculture and domestic purposes. As we have discussed, the relationship between people and their labour, and water are equally important. Hence, it is important to develop, maintain and replenish the water resources on a sustainable basis so that it can serve the present and be available for the future.

Water is the core required for human development. The water is required for the following purposes in our day to day life for the sustainable development:

- **Individual**
 - o Drinking
 - o Cooking
 - o Washing (Cloth, Car, etc.)
 - o Hygiene
 - o Gardening
 - o Home construction etc.
- **Society**
 - o Construction of building
 - o Farming
 - o Maintaining Greenery
 - o Industrial Activities
 - o Medical Services
 - o Recreational Activities etc.
- **National**
 - o Navigation

- o Transportation
- o Agriculture and Food Processing
- o Forestry
- o Corporate
- o Sanitation and Hygiene etc.

Water resources development, optimal utilization and management of water and maintenance is necessary for the sustainability of services from both existing and new developmental activities of all sectors.

2.3 WATER AS A SUSTAINABLE DEVELOPMENT GOAL

As far as the policy debate is concerned, Brundtland Report which was published in 1987 at United Nations discussed the concept of sustainable development and warned of the negative environmental consequences of economic growth and globalization and tried to find possible solutions to problems caused by industrialization and population growth.

It is important to understand that sustainability development means “fulfilling the need of the present without compromising the requirement and capacity of future generation and ensuring the balance between economic growth, care for the environment particularly natural resources and well-being”.

The 2030 Agenda supports the Human Right to Water and Sanitation by aspiring for all: It says that “A world where we reaffirm our commitments regarding the human right to safe drinking water and sanitation and where there is improved hygiene; and where food is sufficient, safe, affordable and nutritious. A world where human habitats are safe, resilient and sustainable and where there is universal access to affordable, reliable and sustainable energy”.

The main focus of the Sustainable Development Goals (SDGs) is to provide an improved water resource that involves building overhead tanks, tube wells, or toilets for drinking and domestic water supply at micro level. The development of mega structures like dams and embankments at the macro level to supply water for irrigation to support agriculture/food production.

Sustainable Development Goal 6 (SGD-6) to ‘ensure availability and sustainable management of water and sanitation for all’ requires explicit attention to gender equality and social inclusion. Universal access to safely managed water, sanitation and hygiene (WASH) and appropriate management of water resources will only be achieved if the rights of women and marginalised people are fulfilled. Inequality, discrimination and social exclusion can be found within water governance and WASH policies, strategies and access to services.

2.4 WATER: ISSUES AND CHALLENGES

Water security is a big issue and has direct impact on sustenance of human life. It is a multidimensional challenge with complex undertones, as water security is both an increasing concern at national and international level in India and critical for sustainable development.

India is facing the water security challenges at international level and potentially affected by the water related actions of upstream countries such as China and Nepal as well as downstream countries Pakistan and Bangladesh on the Indus and the Ganges, respectively.

In view of major issues and challenges, it is relevant to point out that Per capita water availability has shown an unswerving declining trend over the years in the country. We have already discussed the trends of water availability in section 2.2.

As you know, Indian governance model is based on a federal system and subjects are identified for Centre and State to take policy decisions. Presently, water is listed under state list and various state governments take a call on water issues from time to time. When we are discussing issues and challenges in our country from the development point of view, it is important to take note that almost all states are stuck in legal battles for water. You must have seen long pending issue between Karnataka and Tamil Nadu in the form of Cauvery water dispute. Delhi, Haryana and Uttar Pradesh have to target each other every year for the sake of water. The situation in Gujarat and Maharashtra and most other states is not great either. You will have to sort out this policy paralysis to overcome the water crisis in India. Some experts have argued that water be included in the concurrent list so that both Centre and State can take decision on water issues in a harmonious and progressive way.

Total geographical area of India is about 329 m ha which is 2.4% of the world's land area. Only 4% share of the world's fresh water is available in the country to support nearly 15% of the world's population. The main source of water is precipitation that may be rainfall or snowfall. Water cycle or hydrologic cycle is the entire unending process of circulation and redistribution of water by the atmosphere and the earth.

Rainfall is a major source of freshwater in the country. In India, rainfall is highly unpredictable resulting in recurring floods and droughts. Variation of rainfall throughout the country is very high with space and time.

India has monsoon type of climate where more than 70% of rainfall is received during three months (May to September).

The country's main water input from the atmosphere in the form of precipitation is governed by the south-west (summer) and the north-east (winter) monsoons. More than 70% of water input is received in the form of rainfall during four months (May to September). Average annual precipitation is highly variable with space and from about 300 mm in the state of Rajasthan to over 3,000 mm in Meghalaya, Arunachal Pradesh, and Kerala, with a national average of 1,170 mm (Ministry of Water Resources, 1999).

The country has a good network of more than 20 major rivers with several tributaries, out of these some are perennial and some are seasonal. Major rivers like Ganga, Brahmaputra and Indus originated from the Himalayas carrying water throughout the year due to melting of snow and ice which contributes to the base flow during the lean season. The groundwater is an important source of water for our daily life and plays an important role in drinking, irrigation, industrial uses, etc. It accounts for about 80% of domestic water requirement and more than 45% of the total irrigation in the country.

It is important that surface and ground water resources can be managed by adopting water harvesting and conservation measures like rainwater harvesting, preventing pollution with the active involvement of local people that play a significant role in management of this important natural resource.

2.4.1 Water Budgeting

The water is in continuous circulatory movement which is never ending transformation from one state to another on earth as surface water in stream, lakes and oceans or subsurface water (ground water). A water budget is used to quantify the flow of water in and out of a system i.e. a catchment area or drainage area/basin or watershed. It is a hydrological tool. It can also be defined as accounting of all water stored and exchanged on the land surface (rivers, lakes), subsurface (aquifer, groundwater), and atmosphere (precipitation, evaporation).

As understood in the case of economic affairs, experts have started to talk about the planning and water as well. India like many other countries in the world is facing acute water stress and this has jeopardized the entire development spectrum. It is highly important that we need to engage the question of water in the most meaningful manner through the concept of water budgets.

On an average, annually country receives 400-million-hectare meter (mha-m) of precipitation out of which about 135 mha-m is available on the surface and the remaining recharges groundwater. As per the Ministry of Water Resources, GoI (2000), the annual average water resource potential of the country is 1869 billion cubic meters. The utilizable water resource potential of 1123 BCM comprises other 690 BCM surface water and 396 BCM groundwater. In India, rainfall is highly erratic with space and time throughout the country leading to high variation in annual per capita water availability in different parts of the country. Annual per capita water availability is very high in some parts of the country whereas is very limited in some parts for example water availability is as high as 14057 m³ in Brahmaputra basin in Assam and as low as 307 m³ in Sabarmati basin in Gujarat.

Exercise

Prepare an area/village model in which all aspects of water budgeting are discussed under the communication development programme.

2.4.2 Water Usage and Demand

Maximum fresh water is consumed in agriculture to provide life-saving irrigation to the crops. Hence, it is important that both surface and subsurface water should be used judiciously to increase area under irrigation. The share of water for irrigation is likely to be significantly reduced in the years to come because of increase in demand from other sectors. As indicated in Table 2.1, the share of water for agriculture will reduce from 85% in 2000 to 74% in 2050. Present estimated water requirement for industrial water use is around 12 BCM which is likely to increase over fivefold by 2050. The present water demand for domestic use would sharply increase in future due to growing population. Recycling and reuse of water may improve the demand of water.

Table 2.1: Annual water requirement for different uses (in km³)

Use	Year	Year 2010			Year 2025			Year 2050		
	1997-98	Low	High	%	Low	High	%	Low	High	%
Surface Water										
Irrigation	318	330	339	48	325	366	43	375	463	39
Domestic	17	23	24	3	30	36	5	48	65	6
Industries	21	26	26	4	47	47	6	57	57	5
Power	7	14	15	2	25	26	3	50	56	5
Inland navigation	-	7	7	1	10	10	1	15	15	1
Environment-Ecology	-	5	5	1	10	10	1	20	20	2
Evaporation Losses	36	42	42	6	50	50	6	76	76	6
Total	399	447	458	65	497	545	65	641	752	64
Groundwater										
Irrigation	206	213	218	31	236	245	29	253	344	29
Domestic	13	19	19	2	25	26	3	42	46	4
Industries	9	11	11	1	20	20	2	24	24	2
Power	2	4	4	1	6	7	1	13	14	1
Total	230	247	252	35	287	298	35	332	428	36
Total Water Use										
Irrigation	524	543	557	78	561	611	72	628	807	68
Domestic	30	42	43	6	52	62	7	90	111	9
Industries	30	37	37	5	67	67	8	81	81	7
Power	9	18	19	3	31	33	4	63	70	6
Inland navigation	0	7	7	1	10	10	1	15	15	1
Environment-Ecology	0	5	5	1	10	10	1	20	20	2
Evaporation Losses	36	42	42	6	50	50	6	76	76	7
Total	629	694	710	100	784	843	100	973	1180	100

Sources: MoWR (1999)

2.4.3 Significance of Rainwater Harvesting and Conservation

Rainwater harvesting is not only important for fulfilling the ever-increasing water requirement of various sectors but also sustaining water resources for long. Rainwater Harvesting refers to the process of collecting rainwater and storing it for efficient utilization and conservation to control its runoff, evaporation and seepage. Rainwater harvesting is the method to harvest, conserve and utilize the excess rainwater in the best possible manner. The excess rainwater may be used to recharge groundwater by modifying the natural movement of surface water or

through rooftop rainwater harvesting. The basic purpose of artificial groundwater recharge is to replenish depleted groundwater.

Rainwater harvesting in Ancient India

The utilizable surface and groundwater availability in the country is not adequate to meet even the present water requirement of different sectors like public health, industry and agriculture. Rainwater harvest for direct use and groundwater recharge is a common practice since ancient times. A large number of traditional methods of rainwater harvesting are still in use in different parts of the country particularly in arid regions. Today, a number of cities are facing severe water crises particularly lack of availability of safe drinking water and the scenario is even more dismal in rural areas. Water requirement is likely to further increase in future due to constant increase in population, urbanization, industrial growth and change in lifestyle. In order to fulfill the demand of increased food production in the country, the water availability will have to be stepped up. The water scarcity is going to pose a serious threat to the survival of humankind if remedial steps are not taken to ensure reliable water supply. This clearly illustrates the relevance and importance of rainwater harvesting to enhance water availability and recharge depleting groundwaters in the country to meet water demand of different sectors.

There are several advantages of rainwater harvesting, which are as follows:

- free storage space for artificial groundwater recharging;
- evaporation losses are negligible;
- groundwater quality improves by dilution of salts;
- no pollution and high biological purity;
- no adverse social impacts such as displacement of population;
- less runoff and controls soil & water erosion;
- environmental friendly process;
- water stored underground is immune to natural catastrophes;
- controls flood and mitigates effects of drought; and
- increases productivity of aquifers.

2.4.4 Water Quality Deterioration

Surface and subsurface water quality is deteriorated due to water pollution which is posing a serious threat to the environment, human & animal health all over the world particularly in developing countries. Pollution is the by-product of misuse of resources and developmental activities. Our earlier civilizations not only lived along the rivers but also flourished and prospered because people were taking care of these natural resources too. During ancient times, surface water bodies and groundwater were far cleaner and uncontaminated. However, the modern civilization is reversing the trends due to change in life style, fast industrial development and increase in human population. The water bodies are getting polluted and contaminated by untreated sewage water, leaching of agrochemicals from the agricultural fields, industrial effluents etc. The situation has worsened to the extent that almost 70 per cent of rivers and streams not only in our country but also all over the world contain polluted water.

It is necessary to develop water saving technologies and methods which can help in preventing water pollution. These technologies and methods include

rainwater harvesting, water recycling and reuse, and conjunctive use of water for sustaining water supply in long run.

Have we left the water far behind

Right from the second half of the 19th century and early 21st century, we have started increasingly discussing the idea of growth and development in the context of urbanization and industrialization. Everyone is fascinated by the idea of smart cities, but it is very important that we underline the significance of water in the overall concept of development. It is important to look back and see how village life used to be self-sufficient in terms of the water needs and it was very sustainable also from the environmental point of view. When we are discussing the water issue in the context of Human Development, it is imperative to look back and adopt the simple and effective ways of water conservation which were followed by people since time immemorial. Every village used to conserve water through various channels such as, johars and ponds. That was a great lesson in rainwater harvesting, conservation and utilization. The way sedimented mud was used for house renovation and also addressed the issue of desiltation every year in the village set up can be an instructive message for all of us as to how to connect to nature in the most simple and effective way. Dear learners, we are trying to connect to you not to discuss the academic aspects of water alone but to argue to look at the water issue as one of the core issues of human life and do something special in your day-to-day life to address the long-standing issues that we are continuing to face about water.

Check Your Progress 1

Note: 1) Use the space below for your answers.

2) Compare your answers with those given at the end of this Unit.

1) Justify in your own word the importance of water in human life.

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2) Why per capita water availability is decreasing, describe the reasons?

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2.5 MAJOR WATER GOVERNMENT BY GOI

In order to overcome and mitigate the water scarcity stress, governments are taking remedial measures at the national level and also in collaboration with multi-government organizations such as SAARC and UN. We are going to discuss briefly some of the initiatives which have got popular attention to share their impact particularly from the Indian Perspective.

It is relevant to discuss at this point of time that Indian water management and conservation started during the colonial phase and several schemes were initiated

which have strengthened the water security in the country. Prime Minister Sh. Narendra Modi in 2016 highlighted the role of contribution out that a Dr. Ambedkar creation of two powerful technical organizations, namely the Central Waterways, Irrigation and Navigation Commission (CWINC) and the Central Technical Power Board which were the forerunners of the present-day Central Water Commission and Central Electricity Authority. He emphasized that Dr. Ambedkar's pioneering role in securing the water resources in our country cannot be understated.

In the independent India, the central Water and Power Commission was put in place during the period 1952-56 which initiated various plans and schemes for groundwater conservation, irrigation development and drinking water. Indian government has adopted all major international Treaties and Conventions announced by UN agencies from time to time. National Rural Drinking Water Programme (NRDWP) was launched in 2009 with an aim to provide safe and adequate water for drinking, cooking and other domestic needs to every rural person on a sustainable basis. It aimed to provide all rural habitations, government schools, and anganwadis access to safe drinking water.

As you must have noticed, India is microcosm of the whole world which has coastal areas on one hand and deserts on the other. Therefore, the nation requires a comprehensive plan and strategy to ensure seamless water supply for its large-scale population and agricultural needs. To address this need, government of India has initiated Namami Gange Programme which is an Integrated Conservation Mission, approved as 'Flagship Programme' by the Union Government in June 2014 with budget outlay of Rs.20,000 Crore to accomplish the twin objectives of effective abatement of pollution, conservation and rejuvenation of National River Ganga. The programme seeks to cater to the medium and long-term needs of the country with a focused approach on sewerage Treatment Infrastructure, River-Front Development, River Surface Cleaning, Biodiversity, Afforestation, Public Awareness, Industrial Effluent Monitoring etc.

Groundwater is depleting at an alarming rate in many states like Rajasthan, Gujarat, Tamil Nadu, Punjab, Delhi and Haryana. As a consequence, about 60 and 40% blocks in Punjab and Haryana, respectively have turned dark due to overexploitation of groundwater. The term 'dark' indicates that no further groundwater development should be undertaken as the groundwater recharge is considerably less than the groundwater withdrawal.

We need to focus more at utilizing the rainwater to match our water needs. Keeping this in view, Government of India has launched several initiatives and announced subsidized packages for rainwater harvesting. The recent and the most noticeable campaign is Catch the Rain campaign by government of India.

When you are reading and analyzing these campaigns it will be important to add your own individual set of ideas into these campaigns and create local success stories. The development communication will enable you to communicate the government policies in the most effective manner so that the real outcomes of those policies can be achieved. We should support in effective implementation of Govt. initiatives in the water conservation and management.

Therefore, when we are studying the schemes and initiatives on rainwater harvesting, conservation, and utilization, it is imperative that you come out with a strategy to link the policies announced by the Union and State Government to the common man.

When we are discussing policies and initiatives, it is important to understand that the spirit of these policies has to be taken to the masses at large. If you take a stock of water related programs and their success, several experts have argued that we have failed to generate a massive public understanding and sensitization about water issues among common people at large.

It is all the more important now that we are facing such a severe water crisis that we reach out to each and every citizen of the country and impress upon the need of water conservation. Some experts have pointed out that water conservation has to be made a part of Indian educational system right from the school curriculum so that our next generation is trained in the mission to save and conserve water on a day-to-day basis.

We have seen many success stories in the past. So many Indians have made big name for the water struggle. One such example is that of Shri Rajendra Prasad who created an almost impossible narrative of water conservation in the state of Rajasthan. We need to disseminate and propagate the idea of Rainwater harvesting, water conservation and its optimal utilization at all levels through the mode of documentaries, movies and all other available channels. It is expected that after doing this course, you will make a prime mission of it and propagate the idea of water conservation so that a truly great country like India can achieve its true place in the Global hierarchy.

2.6 WATER AND AGRICULTURE

Water is a very important input for agricultural production and plays a critical role in food security. Irrigation has always found an important place in the overall development of agriculture. The rulers at different times in history accorded a very high priority to develop indigenous methods of irrigation. It is clear from the ancient literature that all civilization has taken place around the water source. A number of references are available related to the use of water for irrigation. A system of irrigation was started in Egypt and Mesopotamia during 6000 BC. First time irrigation was started by diverting water from Nile River in Egypt which was flooded every year to the agriculture field to grow crops. During 4000 BC, irrigation practices were adopted judiciously by the farmers.

In India, during 1325-1388 A.D., Western Yamuna canal carrying water from Tajewala Head (between Yamuna Nagar and Paonta Sahib) was constructed for irrigation development in Haryana which was later renovated during 1556-1605 A.D. A large number of canals were constructed during 16th century out of which Upper Bari Doab Canal was popular. In Uttar Pradesh, Eastern Yamuna Canal was constructed to irrigate large tracts during the 18th century. During British rule, the Godavari Delta System and the Krishna delta were developed in 1846 and 1851, respectively.

Till 1951, the irrigation potential was nearly 22.6 M hectare (1 ha= 10000 m², 1 M ha = 106 ha). After independence, Major irrigation projects such as Bhakra Nangal, Gandak, Hirakund, Nagarjunasagar, Chambal, Kosi, Damodar Sardar Sarovar etc. were undertaken to enhance irrigation potential. The significant efforts were also made to enhance groundwater potential. In the southern part of the country, tank irrigation has played a very important role in irrigation.

¹<https://www.forwardpress.in/2020/04/dr-ambedkar-pioneer-of-national-water-resources-development-policy/>

The government has given considerable emphasis during different Five-Year Plans for development of surface and subsurface water potential. At the end of XI Plan (2012), about 119 M ha was realized. As a result of this, Agricultural production has registered a phenomenal increase to meet the requirement of growing population. It has been experienced that the major irrigation projects involve huge construction costs, loss of large areas of productive lands and rehabilitation/ settlement of large numbers of displaced people. It is therefore advisable to undertake minor irrigation projects covering relatively small areas which should be managed by the farmers/ farmers' societies.

In spite of the government's significant weightage on harnessing all replenishable surface and subsurface water resources, more than half of the cultivated area is still dependent on rain. Rainfed agriculture is characterized by low levels of productivity due to low input use. At present, only one third of the cultivated area under irrigation contributes about 55% of total food grain production, whereas two third of rainfed area accounts for only 45% of the total food grain production. Surface and groundwater play a very important role in agriculture particularly in irrigated agriculture.

Groundwater plays a major role in irrigation and domestic water supply. In years of drought, groundwater supplies assured irrigation for stabilizing crop growth during acute shortage of water in surface reservoirs. In rural areas, a large number of wells/tube wells under minor irrigation projects were laid to utilize the groundwater for domestic as well as agriculture purposes. Tube wells were installed in both public and private sectors. Groundwater offers greater control over the supply of water as compared to other sources of irrigation. Timely availability of groundwater helps in high irrigation efficiency (water saving) and higher crop yields.

2.6.1 Irrigation Methods

Irrigation refers to the artificial application of water to the soil to enable crops to have favourable growth particularly in areas where the amount of rainfall is inadequate. Water is applied to the agriculture crops using different methods of irrigation. The irrigation methods may be classified as surface, sub-surface, sprinkler and drip/trickle irrigation methods.

Surface methods of irrigation broadly include the following:

- i) Flood irrigation
- ii) Border irrigation
- iii) Check Basin Irrigation
- iv) Furrow Irrigation

The irrigation method should suit the local conditions. Selection of irrigation methods mainly depend upon the crop, soil, topography, climatic conditions, source of water, water quality, power availability, cost etc. All methods have their advantages and disadvantages. The irrigation method is selected in such a way that applied irrigation water is distributed uniformly throughout the field so that the required quantity of water reaches each plant and water losses are minimal.

The surface irrigation methods are the oldest methods of water application. These are most commonly used because of simplicity and less cost involved. Water

moves over and across the land by gravity flow in surface methods of irrigation. These methods are suitable for soils with low to moderate rate of infiltration and levelled lands with slope less than 2-3 per cent. The main disadvantages of these methods are labour intensive and having less water application efficiencies. The irrigation efficiency in surface irrigation methods is only around 40 per cent. Irrigation efficiencies under different methods of irrigation are presented in Table 2.2.

Table 2.2: Irrigation efficiencies under different methods of irrigation

Irrigation methods	Irrigation efficiencies (%)		
	Conveyance	Application	Overall
Flood	40-50 (Canal) 60-70 (Well)	60-70	30-35
Sprinkler	100	70-80	50-60
Drip/trickle	100	90	80-90

Source: Sivanappan and Padmakumari (1980)

2.6.2 Advance Irrigation Methods for Water Conservation

It is relevant to discuss water conservation and utilization through the modern ways of agricultural practices. Indian farmers have shown great dynamism in embracing the modern-day practices but experts have argued that more area need to be covered in this regard. The following are the advance and efficient methods of irrigation:

- i) Sprinkler irrigation
- ii) Drip irrigation

i) Sprinkler irrigation

In sprinkler irrigation, water is applied through the network of pipes under pressure and distributed to the field in the form of spray, somewhat similar to rain (Fig. 2.2). This method is useful for sandy soils as they have a low water storage capacity, high infiltration rate and absorb water too fast. Hence, they require a small quantity of water in frequent application particularly to the sandy soil with shallow depth. This method can be suitable for uneven topography and where erosion hazards are great. The water is applied uniformly as per the requirement in the field with high irrigation efficiency. However, the energy requirement is high and the method is not suitable for areas having high wind velocity. Initial investment and operating cost of this method is high.



Fig. 2.2: Sprinkler irrigation

i) Drip Irrigation

Drip irrigation method is the most advanced method of irrigation in which water is applied drop by drop with low pressure in the vicinity of the crop root zone. It is also known as trickle irrigation (Fig. 2.3).

In this method, water is applied uniformly and efficiently. If managed properly, evaporation and runoff are minimized. This method is suitable for the areas of water scarcity problem. Fruit crops like grapes, banana, ber, citrus etc., vegetable crops like tomato, brinjal, cauliflower, cabbage etc. and other crops such as coconut, sugarcane, cotton, maize, plantation crop etc. can be irrigated using this method. The water is applied only in the plant roots and only a part of the area is wetted resulted in considerable water saving. The water application efficiency is highest around 90 per cent resulting in 50-100 per cent water saving as compared to surface method of irrigation. However, initial investment is high, the energy is required to run the system and skilled manpower is required to maintain the system



Fig. 2.3: Drip irrigation

Do you know the importance of irrigation efficiency and Irrigation Intensity?

First, we will discuss Irrigation Efficiency.

Irrigation efficiency refers to the ratio of the actual amount of water utilized by the crops to the amount of water applied from a given source. In India, irrigation efficiency is of great concern as it is significantly low (35-40 per cent).

The irrigation intensity can be defined as the ratio of the actual area irrigated to the total culturable command area from a given outlet in a year. Irrigation intensity in the country is only 135 per cent which needs to be considerably increased.

There is a need to increase the irrigation intensity and irrigation efficiency considerably for achieving full advantage of irrigation. The on-farm water management practices like land levelling, irrigation scheduling, method of irrigation depending on type of soil, crop, irrigation source and climate and conjunctive use of surface and groundwater are vital to achieve higher irrigation efficiency. This will help in expanding the irrigation command and increase in production.

2.7 WATER FOR PEOPLE'S WELL BEING

As we have already discussed, water is required for human survival. Thus, access to clean and safe drinking water is one of the important factors of well-being. A comparison of different levels of well-being can be easily visible with the kinds of water access. There are various factors which affect the access of water. The

access of water depends on the situation in which individuals live and obtain water. Water access influences human health, education, life expectancy, well-being and social development. It is a well-known fact that people mainly women and girls walk for long distances to collect water in developing countries particularly in rural areas. In rural areas, lots of time and energy is consumed particularly for women and girl children in collection of water even for domestic purposes. If this time and energy can be saved, it could be used by children for their education. Therefore, reducing the time wasted could improve their subjective well-being, mostly in the case of women, who are the ones who shoulder most of the burden of water collection (UNDP, 2006).

Water, after air is the most fundamental requirement for human survival. But, till now, it has not explicitly recognized as a fundamental human right in India. However, it is true of the Indian legal regime that the recognition of water as a basic right has been indirect, flowing from other rights.

A country can accomplish the goal of development if their people are happy. The main objective of happiness is that each and every citizen of the country will be happy. The scientific study of happiness has been confined to the fields of psychology and sociology. In economics point of view, happiness has traditionally been identified in terms of individual income.

Happiness, subjective well-being or life satisfaction is the scientific term that refers to people's evaluation of their potential to feel happy or not, their current satisfaction with life or their estimation of their own happiness. An individual evaluates his/her level of subjective well-being with regard to his/her circumstances, but also by comparing himself/herself with other people, his/her past experience and his/her future expectations (Frey and Stutzer, 2002).

Water quality is another important concern for the well-being of people as it directly affects the health of people. As a consequence of urbanization, industrialization, change in lifestyle led to demographic growth resulting in an increase in water demand almost in all sectors in many areas. The increase in water demand is the main cause of deterioration of water quality particularly in developing countries. In global terms, water quality is not receiving the attention it deserves, although it is a great problem in many developing countries (Biswas, 2001, 2005). In absolute terms, water quality influences happiness as it prevents people from being affected by illnesses such as diarrhoea (WHO and UNICEF, 2005).

2.8 WATER FOR ECONOMIC DEVELOPMENT

Water plays a significant role in our economy and is at the centre of economic and social development. It is a basic need to maintain health, grow food, manage the environment and create jobs. Water is a complex economic good as it is indirectly linked with economic development of any institution/country. There is a close link between water and economic growth as it permeates all aspects of life on Earth. Water provides essential services for human health, livelihoods and well-being and contributes to economic growth and sustainability of ecosystems. Water is at the centre of economic and social development. There is a direct effect of water on basic human well being and productivity. Water contributes to a number of productive processes that may include direct manufacturing operations in industries or food production activity in agriculture as an element of basic economic activities of any country. Jobs opportunities in the water sector are relatively less. However, there is a good potential of growth for employment in the water sector, but it varies from region to region.

India is an agrarian country and agriculture is the backbone of the country's economy. Agriculture cannot be possible without water. Agriculture is the largest consumer of fresh water at global level. Irrigation consumes almost 70 per cent of the fresh water. Thus, the economy of the country is indirectly with water. In water scarce regions, people are travelling for long distances for water or pumping water from large depths. In both cases, they are paying huge amounts either in the form of labour or electricity or the network of water distribution systems.

Managing water as an economic good is an important way of achieving efficient and equitable economic growth. Better access to clean water, sanitation services and water management creates a tremendous strategy for economic growth of the country. Good management of water resources brings more certainty and efficiency in productivity across economic sectors and contributes to the health of the nation. India like many other parts of the world is in the process of implementing market based water supply reforms to attain equity and efficiency. In India, investment in water resources management, multipurpose hydraulic infrastructure and irrigation have contributed significantly in economic growth. Community water supply takes priority over other water purposes worldwide. Investment in water and sanitation systems in developing economics brings a multitude of economic and social benefits. Water is very important in our national economic welfare.

Check Your Progress 2

Note: 1) Use the space below for your answers.

2) Compare your answers with those given at the end of this Unit.

1) Describe the objectives of Namami Gange Programme.

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2) Explain the importance of advance agricultural practices in water saving.

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3) How is water related to the well-being of people? Justify in your own words.

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4) Describe the importance of water in economic development.

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2.9 LET US SUM UP

Water is perhaps the most important natural resource required for survival of life on the earth. We have discussed in detail about how water holds a crucial role in the course of Human Development with the discussion on all associated issues. In the course of discussion, we have taken up the various issues and challenges with the perspective to offer solutions. We have tried to share the information related to the water usage and demand and diverse ways of water harvesting, conservation and utilization. Since, the Indian economy depends on agriculture and allied activities which consumes a big share of water, we have discussed how modern ways of agriculture practices can be used to minimize input and maximize output. In a nutshell, we have tried to underline the importance of water for the People's well being besides its role in economic development. Hope you have enjoyed studying the unit and you will make most out of it in your day to day life also.

2.10 KEY WORDS

Rainwater harvesting : It is a practice of harvesting, collecting and storing rainwater

Renewable groundwater resource: To restore the groundwater to the original level of availability

Undertone : A feeling, quality or meaning that is not expressed in a direct way

Water scarcity: When per capita annual water availability is less than 1000 m³, it is considered to be a condition of water scarcity.

Water stress: When per capita annual water availability is around 1700 m³, it is considered to be a condition of water stress.

Water table : Water level measured from the ground surface.

2.11 FURTHER READINGS

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2.12 CHECK YOUR PROGRESS: POSSIBLE ANSWERS

Check Your Progress 1

- 1) Refer Section 2 and 2.2.
- 2) Refer Section 2.4.

Check Your Progress 2

- 1) Refer Section 2.5.
- 2) Refer Section 2.6.2.
- 3) Refer Section 2.7.
- 4) Refer Section 2.8

UNIT 3 CHANGING ENVIRONMENT AND ITS IMPACTS

Structure

- 3.0 Introduction
- 3.1 Learning Outcomes
- 3.2 Environment and Our Relationship
 - 3.2.1 Environmental Ethics and its Approaches
 - 3.2.2 Environmental Attitudes
 - 3.2.3 Environmental Justice
 - 3.2.4 Environment and Human Rights
- 3.3 Pristine Environment
- 3.4 Environmental Degradation
 - 3.4.1 Causes of Environmental Degradation
 - 3.4.2 Effects of Environmental Degradation
- 3.5 Global Warming and Climate Change
- 3.6 The Changing Impact
 - 3.6.1 Population growth and Consumption
 - 3.6.2 Issues of Equity
- 3.7 Let Us Sum Up
- 3.8 Key Words
- 3.9 References and Further Readings
- 3.10 Check Your Progress: Possible Answer

3.0 INTRODUCTION

Man always had a synergetic relationship with nature. However, with the development of human civilization, this relationship has changed a lot. Human beings fulfil their needs taking the resources from the planet and also make efforts to renew the economically important resources. The pristine environment has changed over the last decades to such an extent that the healthy survival of human beings and other organisms has become very difficult. Today, we need to understand the worth of precious natural resources and nature's unique services and functions. Due to unsustainable model of development, the environment is degraded and vital resources are becoming scarce. The sensitive species of organisms have become extinct and many are on the verge of extinction. Human beings are facing severe health issues on account of changed climatic conditions and environmental pollution. Therefore it is our prime responsibility to see that the earth remains liveable not only for us but also for the other living organisms. This idea is reflected by the concept of environmental ethics which is the moral responsibility of the individuals to act according to existing information and abilities for the prosperity of the society and community groups.

It would be interesting for you to know that the rising awareness about the issues related to the environment dates back to early nineties. The rise of Environmentalism which emerged as a philosophy and socially driven movement for the protection of nature brought environmental issues to the forefront of discussions and deliberation at this time.

The Rachel Carson's Book "Silent Spring" was a triggering case when the alarming effects of the pesticides were studied, researched and made known to the public. It kick started the modern environmental movement. Likewise, the First Earth Day was celebrated on April 22, 1970, when 20 million Americans walked to the streets to protest environmental degradation and demanded solutions for a cleaner planet.

In India also several movements have taken place since ancient times. The Chipko or the tree hugging movement of 1973 was a forest conservation movement in India. It went on to become a uniting force for many upcoming environmental movements all over the world. Similar kind of eco-awareness also grew in other parts of the World when environmental pollution, oil spills, growing population and accidents took place.

International events such as The UN Conference on Human Environment, 1972, The Earth Summit, 1992, The World Summit on Sustainable Development, 2002 have led to the emergence of global cooperation and recognition of the crucial issues related to sustainability and environment.

Today, the world recognises the environmental consciousness of the teenagers. The 16 year old Environmental activist from Sweden, Greta Thunberg is named "Person of the Year" by Times for her active engagement on the issues of climate change and demanding strict government's action across the world.

In this unit you will learn about the changing environmental conditions and their impact. You will also know about the importance of environmental ethics and attitudes. The unit also focuses on the issues which are related to population and their consumption. In all, the unit explains the causes of changing environmental impact and in turn also categorically discusses about the impact of these changes.

3.1 LEARNING OUTCOMES

After reading this unit, you will be able to:

- Explain the approaches of environmental ethics;
- Explain the causes and effects of environmental degradation;
- Differentiate between weather and climate;
- Describe the mechanism of greenhouse effect; and
- Recognise the importance of equity in sustainable development process.

3.2 ENVIRONMENT AND OUR RELATIONSHIP

The Man-Environment relationship is based on interactions between humans and natural systems. Changes taking place in the environment, feedback mechanisms, the prevailing energy flow, material cycles and systems determine the level of interaction between the species and components of nature. Various approaches and theories have been proposed to explain the dynamic man-environment relationship. Where it is commonly recognised that man's actions influence the environmental settings, this is also true that environment determines human actions.

Man started exploiting the resources recklessly and emitted pollutants and chemicals in the environment at the beginning of the Industrial Age. The aftermath of industrialisation seen in the form of environmental pollution, accidents and deteriorating health of the people aroused the concerns of maintaining the healthy state of the environment. While development was perceived to be the ultimate objective, the importance of environmental ethics and attitude came at the forefront discussions when the governments across the world propagated the importance of environment protection.

3.2.1 Environmental Ethics and its approaches

Ethics is a division of philosophy that determines what actions and attitudes are correct and what are incorrect. In the same way environmental ethics help us to know what actions are right or wrong from the environment point of view. Environmental ethics pertains to our obligation and duties towards environment. It guides us to be careful of our surroundings. Environmental ethics attempts to bring together moral values, ethical behaviour, and appropriate decision making in the best interest of the environment and human population.

The term Environmental Ethics was used for the first time in 1970. It was defined as the part of environmental philosophy which connects the humans and non-humans within the framework of ethics. In other words, the definition of environmental ethics emphasizes equal responsibilities among the various stakeholders in the society. It lays emphasis on the fact that the satisfaction of aesthetic human needs is equally important as provisioning of food and other materials.

Environmental ethics also raises the consideration of appropriate ethics and codes of conduct towards nature and the human's relationship with other life forms. International events like the Earth Summit, Stockholm Declaration, brought the importance of environment issues at and condemned the environmentally destructive activities. Environmental ethics also came to the forefront when there was increased awareness about the impact of population on world's environment. Aldo Leopold quoted in 1949 that "A thing is right when it tends to preserve the integrity, stability, and beauty of the biotic community. It is wrong when it tends otherwise..."

Environmental ethics is the state which establishes the ethical relationship between human beings and environment. In other words it demonstrates the appropriate behaviour of human beings while utilising the services of nature. It helps to describe man's ethical and moral responsibilities towards the environment and considers human values as an important factor. It emphasises that people must make a choice on the basis of the relative importance and accordingly must decide whether to exploit or conserve a particular resource. Human values differ from person to person as every individual does not agree on the same value to nature's components. For example, for poor people, there is no option than burning fuel wood for cooking whereas people who are well off and live in city may consider it as a source of pollution. Thus in this sense environmental ethics are quite comparative and differ from person to person and situation to situation. There are several arguments and counter arguments on the justification of man's interaction with environment. Like, debates over the setting up of hydroelectric plants in spite of knowing the associated environmental changes.

Individuals have different views regarding their obligations concerning the environment.

Some value human needs more than nature and validate the exploitation of environment for the benefit of human beings. This view is known as anthropocentric justification. On the other hand, utilitarian justification believes that conservation of nature is important as it provides individuals with direct economic benefits. This approach highlights that the ultimate goal of ethics should be to do maximum good for the maximum people. For example, mangrove forests are the source of livelihood for the fishermen who depend on aquatic biodiversity. Ecological justification views the ecosystems as the source of the survival of certain species. It is based on accepting the roles of all the species in maintaining ecosystem functions.

There are three primary theories regarding environmental ethics. They are as follows:

Anthropocentrism: Anthropocentrism view considers that human comfort and happiness is utmost important and argues that human beings are the central entities of the universe and so the assumption here is that they can utilize and exploit the environment for their interests and making their lives better. The anthropocentric view is considered to be responsible for the current environmental problems.

Biocentrism: Biocentrism believes that all forms of life are equal and according to Natural rights, they have an inherent right to exist. It is an ethical point of view and applies to all living things. Roots of bio centric ethics lie in our ancient culture and religions. Worshipping of animals and protection of trees has been extensively described in our traditions and historical accounts. However, there are numerous challenges in practising biocentrism.

Eco centrism: Eco centrism is a nature centred system of values unlike anthropocentrism which is basically human centred. It differs from biocentrism in the sense that the abiotic components of environment are also given importance. It believes environment as a whole and humans have moral responsibility of protecting it.

3.2.2 Environmental Attitudes

Environmental attitudes determine our behaviour towards environment which affects its quality. Our attitude determines our action towards preserving or destroying environment.

The three most common attitudes are:

Development attitude- This attitude is anthropocentric by nature. It believes in the supremacy of human beings over all other organisms. Therefore as per this attitude, Earth's resources are only for the benefit of human beings. Present day environmental degradation is the result of the development attitude.

Preservation attitude- This attitude is ecocentric by nature. It believes that nature and its resources are crucial and should be protected at all cost. Advocates of this attitude consider preservation of environment as most important.

Conservation attitude- This attitude seeks to find a balance between unplanned development and preservationist attitude. The proponents of this attitude promote human well-being but also make efforts for environmental protection.

There are several factors which influence Environmental Attitudes such as demographic, social and psychological factors.

3.2.3 Environmental Justice

The foundation of environmental justice was laid down by Robert Bullard, a sociologist who was regarded as the father of environmental justice. In 1991, 17 Principles of Environmental Justice were framed thereafter in 1992 at the Earth Summit; the Environmental justice was discussed at the international platform.

Environmental justice represents impartial and equivalent treatment to all human beings irrespective of their nationality, income level, religion or race while exercising the execution of laws and policies related to the environment. It indicates equal right to all and safe and secure environment to all human beings. You must have seen small children, poor women and elderly working in factories and exposing themselves to toxic chemicals and harmful gases. They work in unsafe conditions and endanger their life due to occupational hazards. These are the examples of environmental injustice. Similarly, people living in disaster prone regions or near polluting industries bear the greatest brunt of environmental pollution.

Environmental justice is thus an important tool to provide clean and healthy living conditions to the masses. It becomes all the more important for the people who have been traditionally living near the polluted sites and have no access to clean water. For instance, one of the environmental justice issues is inadequate access to healthy food. Certain communities in society have no access, availability and affordability to healthy foods. Therefore, impartial distribution of environmental benefits, participation of all the communities in environmental decision-making, giving importance and recognizing the utility of local knowledge and right of individuals to live a decent life all come under the umbrella of environmental justice.

Environmental justice is important for the following reasons:

- i) **Establish Better human relationship with environment** – Our relationship with environment depends on what value we assign to its various parts and subparts. Justice towards nature shapes our thinking and attitude. It is necessary so that we are able to assign value to the nature's services and understand that it is vital for the healthy survival of living beings on this planet. Environmental pollution which threatens the life of living organisms must be condemned. When people understand that environmental justice protects them against nature's vagaries, they tend to protect the environment. Hence a congenial relationship develops between nature and human beings.
- ii) **Conserve Natural Resources** - When people follow environmental justice, it makes them clear that earth's resources are free to be used and every individual has equal access of using them. They realise that unfair distribution and usage of natural resources make them further scarce and limited. Unequal distribution of these resources may lead to resource conflicts and wars.
- iii) **Ensuring long term sustainability**- The basic objective of all conservation measures is to ensure the sustainability of the resources so that the coming generations are also benefitted and are able to enjoy the benefits. Environmental justice strongly supports inter-generational and intra-generational equity and promotes sustainability.

- iv) **Equal distributions of resources** – The natural resources are unevenly distributed and so the rate of extraction also differs. Environmental justice implies fair and equal distribution of resources so that the vital resources such as water, soil, food are made available to all without any distinction.
- v) **Strengthening environmental laws, policies and regulations**- Environmental justice supports the policies and regulations which govern the environment. It supports the principle wherein all individuals and communities are entitled to get support and protection of the Environmental legislations. It treats all people equal while enforcing environmental regulations.

There are pertinent cases wherein the environmental justice is denied to the poor and the vulnerable. We find slums where people are forcibly made to drink filthy water and live in clumsy conditions. Similarly in current times, growing air pollution in Delhi presents a classic example wherein the environmental justice is sought and asked for. The case of the over extraction of ground water for Coca Cola Plant in Uttar Pradesh involved in conflicts with local communities protesting bottling plants. Likewise Dams are persistent sites of conflicts, especially when they are being built and commissioned particularly in Himalayan regions.

Conflicts on the dumping grounds sites like Deonar in Mumbai, Ghazipur in Delhi and many more are increasing as the waste generation manifold and the people living in the nearby areas are getting aware on the severe issues on their health. Similar conflicts have also blown up at construction sites of new airports, highways and other large scale infrastructure projects. The issues are mainly pertinent to their ownership on the land, livelihood and socio cultural problems.

All these conflicts call for environmental justice and protection in the long run. Country's expanding economy is one of the causes of increasing conflicts over environmental issues.

3.2.4 Environment and Human Rights

Human rights are basic rights guaranteed to all human beings irrespective of their religion, place of origin, colour or sex. Right to food, right to education, right to live peacefully, all are examples of human rights. In today's era, when the environment is getting polluted, the need has emerged to declare right to safe environment as a human right. The fundamental right to life is dying out because we are increasingly exposed to harmful chemicals in air, water and even our food. We are forced to consume contaminated water, breathe polluted air and eat food contaminated with pesticides. Environmental conditions regulate the limits of the enjoyment of fundamental human rights. A healthy environment provides healthy living conditions for organisms to thrive. Further, it is understood that human rights cannot be guaranteed in a degraded or polluted environment. We must realize that those who pollute the natural environment are directly or indirectly committing crime against nature and interfering with the enjoyment of human rights.

To address the specific problems in these areas, legal instruments such as laws and rules, dedicated departments and agencies must be strengthened. Earlier, human health, environment and human rights were dealt separately. At the concluding session in Stockholm in 1972, the members recognized that Man is the creature and moulder of environment, which provides opportunities for moral,

social and spiritual growth. The Stockholm Declaration established a foundation linking human rights, human health, and environmental protection

Check Your Progress 1

Note: 1) Use the space below for your answers.

2) Compare your answers with those given at the end of this Unit.

1. Which one of the following reflects environmental justice?

- a) Provisioning of entertainment sources
- b) Availability of roads and railway networks
- c) Providing appropriate living conditions

2. Differentiate between anthropocentrism and ecocentrism.

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3.3 PRISTINE ENVIRONMENT

After having discussed about the ideal man-environment relationship, it is pertinent to recognise the actual present day prevailing conditions. The influence of anthropogenic activities is so much that we have crossed all limits of degrading the parts of environment. The air is polluted with harmful gases and toxins emitted by industries, power plants and vehicles, the water bodies are choked with sewage and plastic, the soil is rich more in contaminants than nutrients. Not only the life of human beings but the survival of other living organisms is in peril. Climate change is emerging as a global emergency and so is the biodiversity loss.

Before we discuss what has changed and what the extent of change is, let us discuss the idea of pure or pristine environment.

The idea of a pristine environment advocates that humans have no place in the natural environment and when human beings start their activities, the environment is no longer pristine. The pure or pristine environment is untouched and without any anthropogenic influence. If the state of the pristine environment has to be maintained or restored, it is important to have a perspective wherein efforts are made to safeguard clean air and fresh water.

The researchers believe, that “previous state or untouched condition” hasn’t existed for thousands of years, so we should make efforts to get back the state of the environment free of pollutants and contaminants. It is so because the pristine or pure environment occurs only in the imagination. It is important to accord a balance between the development activities and environment protection.

3.4 ENVIRONMENTAL DEGRADATION

The state of the environment has changed in a myriad ways since the origin of life on this planet. Various natural as well as anthropogenic factors are responsible

for this. Precisely speaking, environmental degradation is the decline in the quality of the environment through resource depletion, ecosystem destruction, habitat fragmentation and environmental pollution.

Environmental degradation is one of most pertinent environmental issues. Due to the degradation, there are certain ecosystems which never go back to their original state and the flora and fauna inhabiting those places are lost forever. While planning development activities, it must be remembered that ecological destruction and environmental degradation will have far-reaching consequences. The destruction may not always be because of human interference but, we must recognise the extent to which we are damaging the precious resources and valuable services of nature. Environmental responsibility and consideration of environmental ethics play a very important role in this regard.

3.4.1 Causes of Environmental Degradation

There are several factors responsible for degrading the nature of environment. These factors may be natural or man-made.

Natural factors include forest fires, earthquakes, cyclones, volcanic eruptions, landslides and so on. As compared to the natural causes, man-made causes are known to cause environmental degradation at a much wider scale. Some of the important man made causes of environmental degradation are as below:

Deforestation - Irresponsible cutting of trees damages the natural environment and wildlife. To accommodate the increasing demands of the population, large tracts of forests are cleared for agriculture, construction, mining, or other economic activities. Removal of the trees exposes the soil to erosion, kills biodiversity and most importantly alters the climate and precipitation patterns in the region.

Environmental Pollution - Most of the natural environments of the planet has been contaminated and is under huge threat due to the harmful substances and gases emitted from burning of fossil fuel, release of industrial effluents, and use of chemicals in day to day activities. Air pollution, water pollution, soil pollution, etc. poses serious threat to the conditions of environment and also to human health.

Improper Land use Planning: The improper land use planning in the form of unplanned conversion of lands for setting up urban infrastructure, mining, industrial parks, roads, railways, etc. leads to deterioration of natural ecosystems. These activities disturb the geography and terrain of the given region and pose environmental challenges. Conversion of forest land into agricultural fields also comes under improper land use.

Overexploitation of Natural Resources - With increasing population and demand the pressure on current resources is continuously increasing. Due to this, there is enormous amount of extraction of resources. It causes burden on the availability of quality natural resources like water, air, soil, etc. Further, the non-renewable resources are difficult to replenish and it causes scarcity of resources.

Faulty agricultural practices- Intensive agricultural activities degrade the quality of the soil of the agricultural land. It is also water and energy intensive and causes further extraction of resources. The industrial farming uses chemicals

which bring poisons on our table. The “mass production” has replaced the concept of “production by masses” which is detrimental to the agricultural practices in the long run. Run off from agricultural field also causes water pollution.

Generation of Waste Materials - One of the terrible causes of environmental degradation is the generation of large volumes of wastes and trash in huge quantities. Large dumping grounds not only reduce the aesthetic value of a land but also cause land and water pollution. For example, the landfills are responsible for the pollution of soil and underground water. The plants grown on such soils get contaminated and by entering the food chain pose threat to the animals and human beings. The filthy conditions and the stinking landfills and also burning of the wastes makes the places near to these landfill sites unliveable.

3.4.2 Effects of Environmental Degradation

Environmental degradation leads to the following effects on environment and human beings:

Changes in the atmosphere

Environmental degradation disturbs the natural biogeochemical cycles and other activities of nature. Activities like deforestation and mining destroy the natural vegetation and release gases into the atmosphere. It includes particularly global warming and climate change which intensify the occurrence of the climatic natural disasters, and ozone layer depletion. It leads to ecosystem imbalance and loss of vital ecological services.

Decline in Biodiversity

Degradation of the forests and grasslands and other ecosystems poses threat to the survival of species. Habitat fragmentation is considered to be the most important cause of biodiversity loss. Every year, several hundred species are added to the list of threatened and endangered species and many become extinct.

Impact on Human Health

Human health is influenced by environmental degradation. Presence of chemicals in drinking water kills millions of people especially children below 5 years of age. It causes dysentery, cholera and other water borne diseases. Similarly, air pollution kills several million people across the globe and is a leading cause of respiratory disorders including asthma, bronchitis, respiratory infections, etc. The recent studies have indicated that fatalities because of Covid were high at places which were severely polluted.

Poverty

In the majority of developing countries, poverty is attributed to failure of crops, disease infestation, and occurrence of disasters and lack of natural resources which are essential to satisfy basic needs of survival. Poor environment forces people into poverty as there is inadequacy of food and water and the other way round poverty causes environmental pollution. It is a vicious circle where poverty breeds pollution and pollution devoids people of the fundamental right to have clean water and nutritious food.

Apart from the effects mentioned above, environmental degradation also indirectly leads to a number of socio economic consequences.

Check Your Progress 2

Note: 1) Use the space below for your answers.

2) Compare your answers with those given at the end of this Unit.

1 How improper agricultural practices lead to environmental degradation?

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3.5 GLOBAL WARMING AND CLIMATE CHANGE

Global warming is referred to as the increase in temperature of the Earth's surface due to the release of greenhouse gases into the atmosphere. The warming of the Earth has deleterious effects on the climatology, biodiversity and sociology. Before we discuss the effects of Global Warming, let us try to know more about the Green House Effect which is responsible for the increase in temperature of the Earth.

Green House Effect- The greenhouse effect is the phenomenon of trapping of radiations by the earth's atmosphere. Out of the several kinds of radiations emitted by the Sun, some are reflected back, while some are absorbed and re radiated by the gases present in the atmosphere. These gases are methane, carbon dioxide, water vapour, and ozone that absorb and emit infrared radiations and increase the temperature of the earth. These are also known as greenhouse gases. Had there been no greenhouse effect, the temperature of the earth would have been very low and inhabitable for the existence of life forms.

Enhanced Green House Effect: The consequence of enhanced greenhouse effect is the global warming which is happening because of excessive concentration of greenhouse gases particularly carbon-di –oxide and methane.

With the dawn of the industrial revolution, the burning of fossil fuels and electricity generation has led to excess emission of carbon-di oxide into the atmosphere. Due to continuous emission, the percentage of CO₂ in the atmosphere has increased from about 280 parts per million in pre-industrial era to over 409.8 parts per million in 2019. The Green House Effect and Global Warming will become more pronounced as a result of more industrial activity.

Consequences of Global Warming

Global warming has adverse effect on biotic and abiotic components. Increase in land and ocean temperature leads to disruption in the nature's cycles. The warm temperature of the oceans creates a temperature gradient and disturbs the cycle of aquatic organisms. It also leads to expansion of water and compounded by the effect of melting glaciers, the sea level rise. It is a threat to the low lying areas. Rising sea level poses threat in the form of increasing coastal erosion, flooding, and loss of property during any disaster. It is indicated that warmer sea

surface temperatures will cause more frequent and intense tropical storms in the form of hurricanes and cyclones. Coastlines which are already affected by the storms in previous times will have increasing damage to infrastructure.

The rainfall patterns also get disturbed due to temperature increase. Some areas will receive more rainfall and the variations in the pattern of rainfall will also be observed. The duration of rainfall will also change. There will be more rainfall but in less number of days. Increase in temperature will also mean more evaporation and consequently there will be stronger storms. All these are challenging conditions particularly for agriculture and allied sectors. The food availability and access is a concern for future because of the changes in the climatic conditions taking place.

Natural ecosystems will suffer because of changing climate and many species will have to adapt according to the changed habitat conditions. Those which are unable to adapt will eventually be eliminated.

Climate Change

Let us first distinguish between weather and climate.

Weather and climate both the terms are used together to define temperature and atmospheric conditions. The difference between weather and climate is on the basis of time duration. Weather explains the atmospheric conditions over a short period of time, and climate is the atmospheric conditions over an extended period of time. When we discuss climate change, we mean changes in long-term averages of daily weather. We feel that the weather changes several times during a day. On the other hand, climate is the average of weather over time and space. Weather includes sunshine, cloud cover, winds, heat waves, etc. and climate includes patterns of temperature, rainfall and humidity. Climate plays an important role in determining natural ecosystems and socio economic life of the people.

Check Your Progress 3

Note: 1) Use the space below for your answers.

2) Compare your answers with those given at the end of this Unit.

1) What is the difference between global warming and greenhouse effect?

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2) Distinguish between weather and climate.

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3.6 THE CHANGING IMPACT

In the previous sections, we have learnt about the causes and effects of environmental degradation. We have studied that largely human activities are responsible for the current deteriorated state of the environment. It is well understood that environmental degradation destroys the ecology, biodiversity and alters the climatic conditions at any given place. In this section, we will explore the Changing Impact of the Environmental degradation on the socio-economic conditions prevailing in the society.

3.6.1 Population Growth and Consumption

One of the primary causes of the problems of environment has been the rise in population. Due to reasons such as poverty, unemployment and illiteracy, the world population still continues to increase. The world's population increased to six billion people during the 20th century and it may reach 9 billion by 2050. Increase in the population means increase in the demand for food, shelter, energy and other resources for living. It puts pressure on the creation of infrastructure, education and medical facilities. If the population is not adequately fed and educated, it is burden than a resource for any country.

As the world population heads towards 10 billion mark, the requirement of food, energy and other necessities will grow to such an extent that maintaining the resources of the planet will be very difficult. The pressure is on the already scarce resources of the planet. Some resources are dwindling, others are polluted.

The interesting observation has been made today that not only has the more number of people inhabiting the earth poses challenge but also the lifestyle of the population which is a serious concern. With the rise in economic growth and increase in the per capital income of the people, the earth's resources are consumed at an alarming rate. The rich countries of the world are the greatest consumers and it is placed on record that the carbon footprint of the people in the developed countries is much more than the people in the developing countries.

They also produce more waste and as compared to the poorer countries.

When we discuss about the consumption patterns, we need to know what affluence is. The UN conference on the human environment in Stockholm in 1972 identified affluence (i.e., levels of consumption) as the primary source of environmental degradation (United Nations, 1972). In simple words, affluence pertains to abundance and Economic affluence means abundance of wealth. Abundance of wealth leads to increase in purchasing power which is directly proportional to overconsumption.

Over consumption refers to a kind of situation where the resource input has outpaces the sustainable capacity of ecosystem. The consumption patterns of rich and the poor vary greatly. We believe that the consumption habits of the poor and deprived sections of the society are unsustainable because most of the times they indulge in improper land use practices, polluting energy sources and other environmentally degrading practices. Further, overuse of resources such as soil, water, minerals, etc. brings pressure to global commons and puts strain on the carrying capacity of the earth. Likewise, the rich puts pressure on natural resources by consuming more and releasing.

IPAT Equation given by Paul Ehrlich in 1971 explains that the Impact on Environment depends on Population, Affluence and Technology. Therefore, along with controlling unsustainable practices of consumption, it is also essential to reduce the growth of population. In the current times what is conspicuous is that the gap between “people who have” and “people who do not have” is increasing. For example, According to a report, by 2025, 1.8 billion people will be deprived of safe drinking water. Consumption patterns are also influenced by social set ups, technology and urbanization, which often promote consumption by creating new needs and aspirations.

The developed countries comprise of 24 percent of the world’s population, but the share of the consumption ranges from 50 to 90 per cent. In view of this, it is often discussed that with rise in economy, the resources get exhausted more and it may deprive the future generations of the vital resources. The natural resources are fast dwindling and we must understand that today’s decisions and actions are going to affect the future of coming generations. According to the famous saying “We have not inherited the Earth from our fathers. We are borrowing it from our children”, we must see that our actions today do not bring harm to the future generations. We must have concerns for the resource availability to the coming generation. Protection of the earth is everyone’s responsibility and our actions and connections will create a common future for all.

Hence, it is clear that population alone is not responsible for environmental degradation but how people live also is responsible. It is the affluence of the people and their materialistic consumption that influence the overall health of the environment. If a society becomes more affluent, there is increased consumption and use of natural resources. For example, people living in developed countries use more energy, as opposed to people in poor countries.

Let us understand this by the following example:

A person who lives in a village has low ecological footprint than another person who lives in a city and consumes electricity for carrying out day to day activities.

3.6.2 Issues of Equity

The tagline of the UN proposed Sustainable Development Goals is “no one should be left behind.” It implies that equity in the shares of the development is prime concern today. What has been seen since ages is that the fruits of the development are not equally distributed. The poor continue to grow poorer and the rich continue to be rich. The divide between the rich and the poor is widening with the economic development.

The governing bodies and policy makers must see that the marginalized are equally participative in the development process. Environmental risks and rewards must be equally shared among all the sections of the society. Equal access to the services and good must be ensured and there should be no distinction of any kind. The access is mostly determined by economic and power considerations. Prioritising access to public health services, education and creating enabling environment for social development forms the basis for equity in the society.

It is known that in most of the cases, the poor are denied access to safe living and working conditions which makes them more vulnerable to the harmful effects of environmental pollution and degradation. In times of natural disasters, also it

has been seen that the poor, marginalised and the weaker sections of the society suffer the most.

There are a number of challenges in the developing countries which are already struggling and finding out ways to survive and thrive in the current circumstances of resource scarcity, changing climate and polluted environment.

Check Your Progress 4

Note: 1) Use the space below for your answers.

2) Compare your answers with those given at the end of this Unit.

1. What is materialism? How does it influence the consumption of resources?

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3.7 LET US SUM UP

In this unit, we have studied about the changing environment and its impact. We have discussed about the key approaches which determine our behaviour toward the environment. The unit explains the meaning of environmental ethic, environmental justice and throws light on the various approaches of environmental ethics. Environmental degradation takes the form of environmental pollution, decreasing the availability of natural resources, loss of biodiversity and generation of waste. The environmental problems are not region specific as they do not have boundaries. It is necessary to take immediate steps to control emissions and release of chemicals in land, air and water to prevent environmental degradation.

We have studied that the lifestyle of a person also impacts the consumption of resources of the environment. The future generations must be taken into consideration while devising the development plans and strategies.

3.8 KEY WORDS

Environmental justice:	Equal treatment to all the people in environment related policies, regulations and actions.
Ethics	: The set of moral principles which guide the behaviour of a person.
Human Rights	: Human rights are rights to which all human beings are entitled.
Global warming	: Increase in the average surface temperature of the Earth due to the release of greenhouse gases.

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3.10 CHECK YOUR PROGRESS: POSSIBLE ANSWERS

Check Your Progress: 1

- 1) Your answers must include the following points:
 - c
- 2) Your answers must include the following points:
 - Human beings have right to exploit the resources of nature.
 - Environment is above all under ecocentrism .

Check Your Progress 2

- 1) Your answers must include the following points:
 - Deforestation
 - Use of harmful chemicals
 - Environmental pollution
 - Changes in soil structure
 - Energy intensive

Check Your Progress 3

- 1) Your answers must include the following points:
 - The process by which the incoming radiations from the Earth are radiated back to the surface of the Earth.
 - Increase in temperature of the Earth.
- 2) Day to day changes taking place in atmosphere.
Long term changes in the atmosphere defines climate

Check Your Progress 4

- 1) Consumption of resources and energy, scarcity of natural resources, unsustainable means

UNIT 4 COMMUNICATION SUPPORT FOR RURAL DEVELOPMENT

Structure

- 4.0 Introduction
- 4.1 Learning Outcomes
- 4.2 Communication Support for Rural Development
- 4.3 Communication Problems in Rural Development
- 4.4 Communication Planning
 - 4.4.1 Principles of Communication Planning
 - 4.4.2 Communication Strategy Development
 - 4.4.3 Communication Management Planning
- 4.5 Communication Support
 - 4.5.1 Development Support Communication
- 4.6 Let Us Sum Up
- 4.7 References and Further Readings
- 4.8 Check Your Progress: Possible Answers

4.0 INTRODUCTION

In the context of rural development, communication can be defined as the process by which an idea or innovation or technology is transmitted from source to one or more receivers with an intent to change the behaviour of the receivers. The desired effect may involve alterations of the receiver's cognition, attitude and change in the overt behaviour. This process of definition of communication should be combined with feedback which provides greater equality to the participant in a communication event. Unless the communication mechanism allows for a dialogue between the development agent and the potential rural audience, it is likely that communication would become irrelevant. Therefore, proper planning for communication and communication support services are required to implement rural development programmes effectively.

4.1 LEARNING OUTCOMES

This unit is aimed at familiarising you with the importance of communication in rural development (RD). An attempt has been made to give you an over-view of the communication problems, principles of communication planning, strategy development, management planning and organising development support communication. After going through this unit you should be able to:

- Describe communication support in the context of rural development (RD);
- List the problems and principles of communication planning;
- Identify and explain the steps in communication planning;
- Explain Development Support Communication services (DSC);
- Name the components of Development Support Communication unit; and
- Enumerate the functions of Development Support Communication sub-units.

4.2 COMMUNICATION SUPPORT FOR RURAL DEVELOPEMNT

Development workers require scientific and technical information. It is essential that they should have information support by way of gathering, collating, organising, storing and retrieving information.

In most of the cases, science and technological data is in the form of information. It has to be written in a way that is understandable by the target audience because the development workers require support for technical writing and editing.

To communicate effectively different media are being employed, depending on the socio- psychological nature of the audience. The development workers require support in terms of processing of messages relevant to media selected for communication. Therefore, audio-visual production support is envisaged in the context of rural development.

Besides information retrieval, technical writing and editing, audio-visual production support to rural development, it is also necessary to provide printing, photography, video and mailing services.

4.3 COMMUNICATION PROBLEMS IN RURAL DEVELOPMENT

In India, rural development programmes are being carried out without a systematic and well planned communication support component which should be integrated into its activities. This is not to say, however, that there is no communication activity but most of the rural development programmes are beset with communication problems of different magnitude. Obviously, communication activities are being conducted without a cohesive and planned strategy because of the following problems:

Inadequate data about target group on social, psychological, economic, technical and communication related problems. There is a tendency among the rural development staff to report mostly positive information about the situation of target group and to avoid giving negative or unpleasant information. Thus, there is a lack of feed-forward information reflecting positive and negative feedback which is badly needed for communication personnel to respond to the development needs.

Inadequate quality and quantity of the extension communication workers pose problems to rural development. The staff is less in number although some of them have fairly good education, particularly technical knowledge but they are not trained as effective communicators. At present extension communicators discuss technical matters or technical messages during their meetings but not the communication strategy that goes with the technical messages. This again can be attributed to lack of orientation about the importance of communication.

There is no communication competent that could have functioned as a support unit to the various rural development projects such as agriculture, health, nutrition, family planning education, co-operation etc. Most of these projects are planned and implemented by the line department separately and in a disjointed manner.

Hence there is a lack of coordination and linkages leading to problems of integration in terms of its communication strategies.

4.4 COMMUNICATION PLANNING (CP)

In essence, planning is a human process of defining problems, choosing objectives or goals, organising various ways to accomplish goals and measuring progress towards goals achievement.

4.4.1 Principles of Communication Planning

When we plan a comprehensive communication strategy to support rural development, it should be based on certain sound principles.

Following principles could be considered in communication planning (CP).

- A sound CP is based on the-analysis of the facts in the situation,
- Selection of communication problem based on the needs,
- Determination of objectives and solutions which are workable and offer satisfaction. A good CP has balance with flexibility,
- A sound CP has balance with emphasis, A good CP has a definite plan of work,
- Communication planning is a continuous process and is a teaching process as well as a coordinating process,
- A sound CP provides for evaluation of results.

To plan a comprehensive communication strategy to support rural development, an adequate and accurate set of baseline data is imperative. This is essential in order to determine the most feasible, efficient and cost-effective communication approach: A communication plan involves strategy planning and management planning. A strategy reflects the problems chosen and the way communication will be used to solve the problem and the latter is concerned with actions to be taken to implement the strategy. The first covers what to do and second how to make it happen. Therefore, communication planning can be divided into two parts. The first is the process of communication strategy development and the second is the process of management planning.

Check Your Progress 1

Note: 1) Use the space below for your answers.

2) Compare your answers with those given at the end of this Unit.

1) List three effects of communication.

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- 2) List five problems of communication in Rural Development.

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- 3) What do you understand by communication planning?

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- 4) List any five principles of communication planning.

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4.4.2 Communication Strategy Development (CSD)

The communication strategy development lays the foundation for communication planning upon which all other activities are built. It involves following seven steps requiring careful consideration of each of them:

- Step – 1 Baseline data gathering and need assessment,
- Step – 2 Formation of communication objectives, goals,
- Step – 3 Planning analysis and strategy development,
- Step – 4 Audience analysis and segmentation,
- Step – 5 Media analysis,
- Step – 6 Message design and development, and
- Step – 7 Formative evaluation.

Step 1 - Baseline Data Gathering anti Need Assessment: The person in charge of communication support requires relevant baseline data in order to develop suitable CSD and in making decisions. Applied social science research data in areas such as religion and ethics, norms and values, leadership structure, exposure to media and: other socio-cultural, socio-economic as well as socio-psychological information on the audience are needed. The data so collected should serve as resource material to aid the CSD and the others for dissemination of materials on

rural development and related programmes and implementation. Medical practices and utilisation, innovative approaches in communicating to particular target audience, message treatment and analysis etc., are the examples of the former. For dissemination, materials included are those that are relevant and of interest to target audiences such as home and family management, health and nutrition, family planning, agricultural practices and on other economic considerations are obtained from various subject matter specialists. These are then processed and recorded and promoted to the target audience.

In the first step, therefore, there are at least four major components that require data collection.

1) Target Audience

- Size and location of the target audience to be reached.
- Socio-economic profile such as age, income, occupation, number of children etc.
- Socio-cultural profile such as religion, language, education, family patterns, traditional belief systems, habits, norms, values etc.,
- Information sources (what channels are available)
- Media habits.

2) Knowledge, Attitude and Practices

- Level of knowledge, attitude, practices of the specific target group regarding the recommended practices.
- Attitude towards programme approaches.

3) Media Inventory and Impact

- Availability and accessibility of the different communication channels.
- Inventory of hardware (media facilities and infrastructure)
- Inventory of software (slides, films, radio, tapes etc.)
- Media profile (leadership, listenership etc.)
- Cost analysis
- Visual and colour perceptions etc.

4) Manpower Resources and Logistics

- Manpower assessment
- Programme infrastructure
- Related service back-up facilities (health, agriculture, education etc.)
- Media production capabilities
- Manpower and communication materials production cost
- Distribution system.

Step 2 - Formation of Communication Objectives or Goals: Systematic communication planning should begin by carefully defining specific objectives in the light of assessment of interest trends or problems, social goals and values and awareness of the broader implications. Four principal questions should be asked to determine the programme objectives:

- Who: who are the specific target audience to be reached?
- Where: where is the specific group located?

- Why: why is the specific group chosen to be a target group?
- What message: what type of content of message should be communicated to this specific target?

The specific objectives should be defined as sharply as possible because only then can we hope to measure or evaluate how well these have been met.

Step 3 - Planning-Analysis and Strategy Development: After identifying the specific communication objectives to be accomplished and the needs gap at the general level of analysis, the next step is to translate these objectives and needs statements into a workable communication strategy. There are at least two inter-related aspects of communication strategy development:

- Choice of communication approaches.
- Identifying the type of messages to be conveyed.

In choosing communication approach or combination of approaches one will have to determine first, what effects s/he wants to get. Depending on the knowledge; attitude and practice level of a particular target-group regarding the recommended idea there are several communication approaches that can be utilised;

- To increase awareness and knowledge level: the information transmission approach.
- To provide or increase motivation: persuasion approach and dialogue approach.
- To provide specific technical knowhow: instruction approach.

It is often the case that those audiences who participate in communication activities are not those whose attitudes need change. Likewise with nutrition, health, agriculture, and education classes, those who attend are themselves well aware of such practices, but the less informed ones do not seem interested. Perhaps, a different communication strategy could be designed for late adopters that is most suitable to their desires and needs. In order to design an effective communication strategy one would need to know the factors that influence and motivate behavioural changes of the target group.

Step 4 - Audience Analysis and Segmentation: Analysis of the target audience is one of the most important factors in designing an effective communication strategy. Due to the different characteristics as well as needs of the target audience; audience segmentation is usually necessary. For each target group, a specific communication strategy is thus required. It is also important to assign priority as to which target group should be reached most urgently.

Step 5 - Media Selection: In selecting a particular media or channel for utilisation one will have to list the communication channels that reach the particular target or audience. Then, media must be evaluated in terms of its applicability to accomplish the specific communication objectives. The communication planners will also have to determine whether the media is available for use as well as the cost effectiveness. Another, important factor to consider is the coordination or integration of multi-media communication activities to reach a particular target group. In essence, it is not sufficient to say “use radio” for this advice needs qualification in many instances, such as, for what audience, under what

circumstances, for what type of message and, with what other media re-enforcement.

Step 6 - Message Design and Development: In this phase the message, theme, treatment and presentation should be decided. The main activity will thus be the designing of the proto-type communication materials which also require formative evaluation such as presenting of the prototype materials to the target audience. The pretesting results can guide the necessary revisions of the proto-type materials before it goes for the final, large scale production process.

Step 7 - Formative Evaluation: It is an evaluation that help communication planners in developing and improving communication approaches and materials. Formative evaluation has to be built-in at all steps of the Communication Strategy Development (CSD) and Communication Management Planning (CMP), as it provides feedback for modifications and helps to take up corrective measures of an ongoing communication programme.

Check Your Progress 2

Note: 1) Use the space below for your answers.

2) Compare your answers with those given at the end of this Unit.

1) What do you understand about the following terms:

a) Communication strategy

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b) Communication management planning

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2) List the steps to be followed in communication strategy development.

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3) List four major components on which data is collected in detail.

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4) Indicate the approach to be followed under the following situation:

To increase awareness and knowledge level

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To motivate

To teach specific technical know-how

4.4.3 Communication Management Planning (CMP)

The basic questions to be answered in Communication Management Planning (CMP) are, who will do what and when. It is also essential to organise a good management information system which can provide planners with routine information on personnel, finance and logistics. As the CMP is the key link between strategy and actions, careful considerations of the following steps is essential:

Management Planning: One of the most difficult problems that communication planners face is that of scheduling and coordinating large numbers of communication activities. Planners must arrange the communication activities in a logical way, estimate the time and resources required to complete the activity, and keep constant watch on the schedule as the activities unfold. There are at least 3 scheduling techniques:

- The monthly work plan
- The time effect chart
- The net-work scheduling (PERT)

Having a well-thought-out communication strategy, scheduling of activities and events of implementation is only a part of the planning process. Resources necessary to accomplish the activities must be budgeted. In establishing these estimates, it is essential to take into account the money and people required to complete activities within the estimate, thus balancing the three resources namely money, people and time and to come up with a mix which can be achieved within the resources available.

Manpower Training: When we design a communication strategy, the persons who will be implementation the strategy must also be ready and prepared to carry out the activities as planned. Therefore, an assessment of available manpower for the tasks to be accomplished is needed. After the selected persons have been chosen to carryout the plan, their tasks and the training needs should be identified and based on this, training for those who need it should be provided. By so doing, each person engaged in the implementation of the plan is prepared and well-equipped to handle the assigned task.

Implementation: At this step, all the necessary activities in achieving the planned communication objectives should be carried out. The personnel involved in the implementation of the communication activities should be constantly informed about the task to be performed and the problems encountered by them should be resolved expeditiously through careful monitoring of the activities.

Summative Evaluation: In progress assessment and decision-making, evaluation is an important means to make rational choice between alternative practices and to validate programme improvement. The purpose of programme evaluation in communication activities is that it serves as:

- an integral element in project management to monitor project performance or effectiveness,
- a device for improving resource allocation and programme management and,
- a means of policy formulation by having regular retrospective examinations of communication activities.

Programme evaluation is thus concerned with developing information to be used in modifying and improving communication strategy. Its focus is not only communication goals and strategies, but in addition, it also helps to analyse effects of outcome of the communication programme in relation to objective, inputs, outputs and cost. The results of summative evaluation should be considered as inputs for formulating new communication objectives and in designing new communication strategies.

Check Your Progress 3

Note: 1) Use the space below for your answers.

2) Compare your answers with those given at the end of this Unit.

- 1) List three scheduling techniques.

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- 2) How is evaluation helpful in communication?

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4.5 COMMUNICATION SUPPORT

Communication support can be broadly defined as an organisation's planned use of information, and communication resources to help achieve its goals. Communication resource include: manpower, money, facilities, equipment, communication materials and media.

Communication support can also be defined as an unit of development organisation whose function is to facilitate communication by providing services such as technical writing and editing, graphic arts, photography, audio-visual production, type setting, printing, information management and retrieval and mailing services. The various components of communication support services can be defined functionally as follows:

Writing: Review and interpret research in the areas relevant to rural development such, as rural development planning and implementation, agriculture, health,

rural engineering, family planning etc.; rewrite to a level understandable by subject matter specialists, extension workers and farmers; it may even include news release.

Editing: Refine manuscript to conform to accepted standards and for a particular purpose, before printing.

Photography: Document research work and events with photographs and production of slides, movie or video films and other photo reproduction.

Visual and Audio-Visual Production: Develop and produce visual, audio-visual instructional materials such as charts, tape-slides and other teaching aids.

Information Management and Retrieval: Acquire research studies and technical information. Store in such a way which can be easily retrieved for dissemination.

Graphic Arts: Design page layouts for publications, produce graphs and illustrations for audio-visual aids and publications.

Printing: Produce multiple copies of printed materials.

Considering the communication planning and support services, an organisational structure could be built around these communication tasks. Such an organisational unit is popularly known as Development Support Communication (DSC) unit. In most of the United Nations Development Programmes, development support communication has become an accepted pattern. Presently, Development Support Communication has become an integral part of the rural development programmes sponsored by the United Nations.

A prototype organisational structure for a DSC unit suggested by the UNDP-DTCP is presented below:

4.5.1 Development Support Communication

DSC Unit Head

1) Programme Section Functions

- Developing DSC Plans with the agency staff
- Consultation with Technical specialists on contents of message
- Information data collection
- Media selection
- Creative design
- Writing scripts
- Pretesting materials & methods
- Monitoring and evaluation.

2) Production Section Functions

- Graphic art work
- Printing
- Photography
- Audio/Video recording
- Mass Production
- Storage of master materials equipment facilities and maintenance

3) Distribution Section Functions

- Inventory and Storage of internal and external DSC materials
- Library of communication and audio-visual materials and references
- Distribution

The DSC unit is concerned with both communication planning and support services. Its activities are primarily directed to the field services and staff training functions related to rural development agencies. The DSC staff cannot create the technical content of the messages in isolation. The content or substance of the programme comes from the agency's, technical or subject-matter specialists. Furthermore, an agency's training and field operation are normally controlled by other agency units.

Thus communication support staff cannot afford to work in isolation. They must plan and execute the DSC programme in close collaboration with the agency's planners, technical specialists, trainers, supervisors, field workers and other relevant staff.

Check Your Progress 4

Note: 1) Use the space below for your answers.

2) Compare your answers with those given at the end of this Unit.

1) What do you understand by the term communication support?

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2) List the communication support services needed for production section.

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4.6 LET US SUM UP

Hitherto, not much attention has been given to communication planning and support services. It is recognised that without proper planning for communication, rural development programmes cannot be implemented effectively. To benefit from rural development programmes people need to be informed, motivated and instructed about the technologies in ways understandable and acceptable to them for communication support plays an important role in establishing dialogue and in getting feedback. Neglect of communication support may spell disaster on the

development effort. We have dealt in detail various aspects of communication support in this unit. We tried to identify communication problems in rural development. Then we shifted to communication planning. Communication planning is a human process. It starts with analysing the situation, identifying communication needs, choosing objectives, organising various ways to accomplish goals and evaluation. The process of planning can be divided into two major parts;

- 1) Communication strategy development involving several steps, and
- 2) Communication management planning.

Communication support is planned information support services, and communication research. The support services include technical writing and editing, type setting, printing, information management and retrieval and mailing services.

Both communication planning and communication support services can be organised in the form of an unit namely Development Support Communication (DSC) unit. The DSC unit consists of a unit head under whom sub-units like communication Programme Section, production Section and Distribution Section could be organised. Each of these sections have definite functions to perform.

4.7 REFERENCES AND FURTHER READINGS

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2. Gecolea H. Romeo, *A Primer on Development Support Communication*. UNDP Asia Pacific Programme for Development Training and Communication Planning, Bangkok, Thailand.
3. Roy, G.L., 1991. *Extension Communication and Management*. Nayar Prakash, '206, Bidhan Sarani, Calcutta, India.

4.8 CHECK YOUR PROGRESS: POSSIBLE ANSWERS

Check Your Progress - 1

1.
 - a) Cognition or knowledge
 - b) Attitudes
 - c) Overt behaviour
2.
 - a) Lack of data on audience and characteristics
 - b) Concentration on positive feedback
 - c) Lack of communication professionals
 - d) Importance given to technical subjects
 - e) Lack of communication support of RD
3. Communication planning is a human process of defining the problem, choosing objectives and accomplishing objectives and evaluation.

4.
 - a) Analysis of situation
 - b) Based on peoples needs
 - c) Clear objectives
 - d) Definite plan of work
 - e) Measurable results

Check Your Progress 2

1. It reflects problems to be solved and the different ways communication is used in solving problems.
2. Action to be taken to implement the strategy
 - a) Baseline data collection and identification of communication needs,
 - b) Formulation of objectives,
 - c) Strategy development,
 - d) Audience, segmentation
 - e) Media analysis
3.
 - a) Target audience
 - b) Knowledge attitudes and practices
 - c) Media inventory
 - d) Manpower resources
4.
 - a) Information transmission approach
 - b) Persuasion approach
 - c) Instruction approach

Check Your Progress 3

1.
 - a) Monthly work plan
 - b) The time effort chart
 - c) The network scheduling
2.
 - a) Monitor project performance
 - b) Resource allocation
 - c) Policy formulation

Check Your Progress 4

1. Communication support includes technical writing, editing, audio-visual production, graphics, type setting and printing, information retrieval and mailing services.
2.
 - a) Graphic art work
 - b) Printing
 - c) Photography
 - d) Audio/Video recording
 - e) Storage of master materials
 - f) Equipments and maintenance.